Performance Analysis and Optimization of RSMA Enabled UAV-Aided IBL and FBL Communication with Imperfect SIC and CSI

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***Abstract*—In this work, we investigate rate-splitting multiple access (RSMA) for a multiuser downlink wireless network consisting of an unmanned aerial vehicle (UAV)-assisted base station (BS) that serves multiple ground users (GUs) simultaneously. Considering two different transmission regimes, namely infinite blocklength (IBL), and finite blocklength (FBL), we analyze the performance of the considered network under the effect of imperfections in channel state information (CSI) estimation and successive interference cancellation (SIC) with the probabilistic line of sight fading channels. For IBL transmission, we derive the closed-form expressions of the outage probability, throughput, and achievable ergodic rate at each GU. Furthermore, for short packet communication, the closed-form expressions of block error rate (BLER), goodput, and achievable ergodic rate are determined with FBL transmission. Moreover, for the FBL regime we also formulate an optimization problem that jointly optimizes the 3D-position of the UAV, power allocated to each user and common rate distribution at each user to maximize the ergodic sum rate subject to the practical constraints such as maximum tolerable BLER, minimum private and total rate at each user. We then propose an alternating optimization-based iterative algorithm to solve the problem. Monte-Carlo simulations are used to verify the accuracy of derived analytical results and demonstrate the trade-off between transmit power and achievable BLER. In addition to this, the effectiveness of RSMA in UAV-assisted communication under IBL and FBL transmission regimes is also observed compared to non-orthogonal multiple access.**

***Index Terms*—Block error rate (BLER), ergodic rate, outage probability, rate-splitting multiple access (RSMA), unmanned aerial vehicle (UAV), infinite blocklength (IBL), finite blocklength (FBL).**

I. Introduction

UNMANNED aerial vehicle (UAV) is a prominent technology for next generation (beyond fifth generation (B5G) and sixth-generation (6G)) wireless networks and will play a vital role in strengthening the Industry 4.0 revolution [1]. Controllable mobility, easy implementation, low cost, user-friendly operations, lightweight, and strong line of sight (LoS) channels are the factors that lead to the popularity of UAVs [2]–[5]. Among several applications, one of the most common uses of the UAV is using it as an aerial base station (BS), called UAV-BS, which provides improved spectral efficiency and better connectivity, and is a potential candidate to assist ”connectivity from the sky” required in B5G and 6G. A strategic difference of UAV-BSs with respect to the conventional BSs is their ability to deploy on-demand networks at specific locations, thanks to UAVs aforementioned exceptional characteristics. This allows ground users (GUs) to have premium services with high quality wireless links, poor degradation, high capacity and low interference [6]–[9].

Further, in order to have efficient resource utilization, the use of non-orthogonal multiple access (NOMA) in UAV-BS aided multiuser communication has been studied extensively by the researchers [10], [11]. It uses superposition coding at the transmitter to superimpose the individual information signals of all users into a single waveform and successive interference cancellation (SIC) is applied that decodes the signals one by one following a preset decoding order until the respective receiver finds its desired signal. However, in order to handle the increase in SIC complexity with increase in the number of users, NOMA communication requires an extremely complex receiver design, which is its major limitation. Indeed, for a K-user broadcast channel (BC), the strongest user needs to decode the K-1 messages of all co-scheduled users using SIC and therefore peel off K-1 layers before accessing its intended stream. This calls for ways to decrease the number of SIC layers at each user. In addition to this, since there exists no natural order for the users' channels in NOMA, the precoders, the groups, and the decoding orders have to be jointly optimized by the scheduler at the transmitter significantly increasing the complexity of the transmitter. To overcome these limitation of the NOMA, a more general and powerful multiple access framework based on linearly precoded rate-splitting multiple access (RSMA) at the transmitter and SIC at the receivers was introduced in [12]. This enables to decode part of the interference and treat the remaining part of the interference as noise.

The dominance of RSMA on the performance of wireless networks over other multiple access techniques such as NOMA, orthogonal multiple access (OMA), etc., was demonstrated in numerous studies [12]–[15]. In RSMA, each user's message is divided into the common message and the private message on the transmitters' side. Therefore, only

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two-step decoding is required at each receiver, resulting in reduced complexity compared to NOMA. In recent literature, researchers have studied the use of RSMA in UAV-assisted wireless networks [16]–[18]. Specifically, the authors in [17] investigated the use of RSMA-based UAV-BS that communicates with GUs in case of breakdown of the main BS in cloud-radio access networks (CRANs). In [18], the integration of RSMA and UAV-BS was studied and a non-convex problem of joint UAV placement and RSMA precoding was solved using an alternative optimization method to maximize the weighted data rate. However, due to the involved analytical complexity, none of the above works in [16]–[18] have analyzed the performance of RSMA-based UAV-assisted cellular network. Therefore, in a similar context, we analyzed the outage probability and achievable throughput in an RSMA-enhanced UAV-BS downlink multiuser network and maximized the sum throughput of the considered network by optimizing the position of the UAV-BS [19].

The aforementioned works [16]-[19] considered idealistic approach and used conventional infinite blocklength (IBL) transmission regime message packets to investigate UAV-aided communication. However, to support ultra reliable and low latency communications (URLLC) as a new service type of B5G and 6G cellular systems, short packet transmissions are essential [20], [21]. In this case, errors cannot be reduced to arbitrarily low for a given coding rate due to the limited packet size. Motivated by this, the authors in [20] have developed a new fundamental framework for short-packet communications and derived an error probability bound for a finite blocklength (FBL) and coding rate. It was shown that the block error rate (BLER) increases as the blocklength of the system decreases. This new theoretical framework opens a new research direction for the revisit of conventional communication networks, which were mainly designed based on the Shannon formula and thus cannot be directly applied to short-packet communications.

It should be noted that in existing studies under the Shannon theory, errors can be avoided when the transmission rate is below the Shannon capacity. Thus, the outage probability becomes an appropriate performance metric to investigate the performance of such systems [19]. However, when it comes to short-packet communication scenarios, there always exists a non-zero error probability even when the transmission rate is below Shannon capacity. According to the work in [20], the achievable rate in short packet communications depends on both the blocklength and the desired BLER. Note that the analysis in [16]-[19] cannot be directly applied to short packet communications in order to meet URLLC requirements in B5G and 6G. Thus, the investigation of the probable performance of such network is extremely important with the short packet transmission under the FBL regime in order to fully understand its ability to obtain the desired quality of service (QoS). In this regard, several studies have investigated the short packet communication in UAV-BS-aided without RSMA and RSMA-enhanced networks with fixed BS [22]–[25]. In [22], the total energy consumed by an FBL transmission-based UAV-BS-aided multiuser wireless network was minimized by jointly optimizing transmit power, UAV-trajectory, and latency. A block coordinate descent-based algorithm was proposed in [23] that jointly allocates the optimal power control and resource allocation in UAV-assisted URLLC networks. Also, the energy efficiency (EE) of a UAV-aided cognitive IoT network with short packet communication was maximized in [24]. In the case of RSMA-enhanced FBL-aided downlink communication, in [25] the sum rate was maximized by designing the optimal linear precoders considering the FBL constraints. It is noteworthy that, although being a futuristic technology, the integration of UAV-BS with the RSMA is still in its elementary stage, thus, lacking an appropriate and suitable study that presents its full capabilities and practical limitations under the FBL regime.

Nevertheless, the aforementioned works [22]–[25] were mainly focused on the optimal resource allocation, therefore, the literature related to the performance evaluation of such systems is relatively scarce. The advantage of UAV in short packet communication was studied in [26] for line of sight (LoS) channel conditions. The authors in [27] discussed the achievable BLER and goodput in multiple UAV-aided communication considering the FBL transmission regime. In [28], the performance of reconfigurable intelligent surface in UAV-aided vehicular communication was investigated for short packet communication, and closed-form expressions of coverage probability and BLER were derived. However, to date, most works have considered only a single user for presenting the analytical framework. Further, the assumptions of perfect channel state information (CSI) and perfect SIC at GU lead to simpler but impractical considerations. Moreover, in order to have a near practical understating, it is essential to consider these imperfections and investigate their impact on the performance of such networks. For example, several works investigated the effect of imperfect CSI and imperfect SIC on the performance of NOMA-assisted communication [29]–[32]. Also, in the case of RSMA-assisted communication, the impact of imperfection in CSI estimations were analyzed in a few works [33]–[35]. RSMA was studied in multi-carrier systems in [33] and optimal resource allocation was achieved using a novel three-step algorithm under the consideration of both perfect and imperfect CSI. In [34], the weighted sum rate in an RSMA-enhanced multi-antenna-assisted downlink network was maximized considering the impact of the imperfection of CSI at the transmitter (CSIT) and CSI at the receiver (CSIR). Similarly, the robustness of RMSA in the presence of CSIT and CSIR was demonstrated in [35] and an alternating optimization-based algorithm was proposed to maximize the sum rate. Further, the impact of imperfection in SIC was discussed in [36] wherein suitable bounds of different power coefficients and the distributed common rate at each user were derived.

A. Motivation and Contribution

As discussed above, the integration of RSMA in UAV-BS is very beneficial for B5G and 6G networks. Besides, RSMA also finds benefits in single-antenna settings (i.e., each user and BS have a single antenna). Indeed, although NOMA is capacity achieving in single-input single-output (SISO) broadcast channel (BC), it is very complex due to

TABLE I: Comparison of this work with existing literature on RSMA-aided communication

	[16]	[17]	[18]	[19]	[22]	[23]	[24]	[25]	[26]	[27]	[28]	[33]	[34]	[35]	[36]	our work
RSMA	✓	✓	✓	✓	X	X	X	✓	X	X	X	✓	✓	✓	✓	✓
Conventional BS	✓	✓	X	X	✓	X	X	✓	✓	X	X	✓	✓	✓	✓	X
UAV-BS	X	X	✓	✓	X	✓	✓	X	X	✓	✓	X	X	X	X	✓
UAV as Relay	X	✓	X	X	✓	X	X	X	✓	X	X	X	X	X	X	X
Sum Rate Maximization	✓	✓	✓	X	X	X	X	✓	✓	X	X	✓	✓	✓	X	✓
Throughput Maximization	X	X	X	✓	X	X	X	X	X	X	X	X	X	X	X	X
UAV-Position Optimization	✓	X	✓	✓	✓	✓	X	X	X	X	✓	X	X	X	X	✓
Variable Pathloss exponent	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	✓
IBL	✓	✓	✓	✓	✓	X	X	X	X	✓	✓	✓	✓	✓	✓	✓
FBL	X	X	X	X	X	✓	✓	✓	✓	✓	✓	X	X	X	X	✓
Imperfect CSI	X	X	X	X	X	X	X	X	X	X	✓	X	✓	✓	X	✓
Imperfect SIC	X	X	X	X	X	X	X	X	X	X	X	X	X	X	✓	✓
Performance Analysis	X	X	X	✓	X	X	X	X	✓	✓	✓	X	X	X	✓	✓
Outage Probability	X	X	X	✓	X	X	X	X	✓	✓	✓	X	X	X	X	✓
Throughput	X	X	X	✓	X	X	X	X	X	✓	✓	X	X	X	X	✓
BLER	X	X	X	X	X	X	X	X	✓	✓	✓	X	X	X	X	✓
Sum Rate	X	X	X	X	X	X	X	X	X	X	X	X	X	X	✓	✓

multi-layer SIC. In contrast, [12] demonstrated that in SISO BC, RSMA can get similar performance to NOMA but with a single SIC, therefore significantly reducing the receiver complexity. This calls for strong benefits in marrying RSMA with UAV-BS even in the presence of single-antenna nodes and analyze the performance metrics such as BLER and achievable rate for short packet communication in RSMA-aided UAV-assisted networks that can guide system designers to attain several practical insights. However, due to insufficient literature, the performance of RSMA in UAV-assisted wireless networks needs to be explored extensively in order to utilize their full potential. In the following, the comprehensive and comparative discussion about the research gaps in the existing literature and, thereafter, the motivations for the current work are presented.

- Most of the work in the literature considered perfect SIC and CSI for their investigation leading to the impractical scenario. Although some of the recent works addressed this but considering either imperfect CSI or imperfect SIC, not both. This article is the first attempt to analyze the impact of imperfection in both, SIC and CSI, on the system performance.
- Also the investigation of any system with an IBL regime is idealistic and unrealistic. However, in order to support reliable and low latency communication in 6G and industry 4.0 in a more realistic scenario, the robustness of any technology should be investigated with the FBL regime. Apart from [25], no other work studied the performance of the RSMA in the FBL regime. Additionally, to the best of the author's knowledge, the study of the FBL regime in the RSMA-enhanced UAV-BS-aided system is still untouched, which is the main focus of this work.
- Unlike the perfect CSI and SIC case, the imperfect CSI and SIC lead to the addition of additional interference and more complex integrals while analyzing the system performance, hence increasing the analytical complexity to many folds.
- Unlike conventional/fixed BS, due involvement of 3D-channel modeling in a UAV-BS aided communication, the optimal UAV placement plays a vital role in achieving better network performance. Additionally, the slight variation in UAV position leads to a significant change in the pathloss (as the channel gain depends on the distance between the UAV and users). Thus, finding an optimal position in UAV-aided communication is extremely important and challenging.
- Due to the involvement of RSMA in an FBL regime and to ensure the user fairness, the constraint of the minimum required rate (common+private) at each user is levied. Additionally, the limited flying zone of the UAV further imposes new challenges of resource management. Therefore, for efficient communication, a significant problem is to provide optimum power allocation to each user at the UAV-BS and common rate distribution for each user, while guaranteeing the minimum QoS requirement for each user under the strict constraints of the limited UAV flying zone, the maximum tolerable BLER and achievable common rate. Consequently, in order to achieve the maximum rate gains from UAV-BS-enabled communication, it is important to jointly optimize multiple relevant parameters, e.g., UAV position, common rate distribution, power allocation and achievable BLER. To the authors' best knowledge, this important problem, within a RSMA setting, has not been treated previously

In order to improve the understanding of this study, a detailed comparison of our work with existing literature on RSMA or UAV-BS aided communication is highlighted in Table I.

Motivated by the aforementioned discussions and considering the fact that UAV-BS will play a pivotal role in the next-generation (i.e., B5G and 6G) wireless networks as a cost, power and spectrum efficient technology, through this work, we aim to provide a benchmark mathematical framework on the impact of RSMA in UAV-BS aided short packet communication using the FBL regime with more realistic assumptions of CSI and SIC estimation errors. Accordingly, we address the major challenges of the integration of UAV-BS with RSMA, and focus on the outage probability, throughput, BLER, goodput, and achievable ergodic rate performance analysis at each user. Therefore, we explore for the first time the interplay between RSMA and UAV-BS considering the effect of CSI and SIC errors and show that RSMA is a promising multiple access strategy for UAV-BS-assisted multiuser communications under both IBL and FBL regimes. *The key contributions of this work are as follows:*

- We investigate a UAV-BS-aided multiuser communication network where UAV-BS uses RSMA signaling for information transmission. By considering two different transmission regimes, namely *IBL transmission and FBL transmission*, the performance of the considered network is analyzed under a probabilistic LoS fading channel.
- In contrast to [16], [17], where the main focus was on the resource allocation and trajectory design in IBL under the perfect CSI and SIC, we analyze the network's performance by deriving the closed-form expressions for the outage probability, throughput, and achievable ergodic rate[1] under the *imperfect CSI and SIC*.
- Unlike IBL transmission [6]-[17], to support reliable and low latency communication in 6G and Industry 4.0, *we consider FBL transmission* and derive the closed-form expressions for the average BLER, goodput, and achievable ergodic sum rate[2] under the realistic assumptions of *imperfect CSI and SIC* for the first time.
- Unlike [19] wherein only the position of the UAV is optimized under an idealized IBL regime with perfect SIC and CSI, we formulate an optimization problem that jointly optimizes the 3D-position of the UAV, power allocated to each user and common rate distribution at each user to maximize the ergodic sum rate for the FBL regime subject to the practical constraints such as maximum tolerable BLER, minimum private and total rate at each user under the imperfect SIC and CSI estimates. Due to the non-convex nature of the formulated problem, we first reformulate the main problem into a much simpler form, then propose an alternating optimization (AO) based iterative algorithm that solves each variable alternatively considering other variables constant until the convergence is achieved.
- The accuracy of derived analytical expressions is validated using Monte-Carlo simulations. Additionally, numerical results are also presented to provide some important and useful insights related to the impact of these imperfections on the performance of the considered network. Further, we discuss the *trade-off between transmit power and achievable BLER*. Finally, we demonstrate the effectiveness and usefulness of RSMA in such scenarios as compared to NOMA.

Notations: $\Gamma(\cdot)$ represents the Gamma function, $\Gamma(\cdot,\cdot)$ and $\gamma(\cdot,\cdot)$ denote the upper and lower incomplete Gamma, respectively. A complex Gaussian distributed random variable is depicted by $\mathcal{CN}(m,\sigma^2)$ having mean m and variance σ^2. probability density function (PDF) and cumulative density function (CDF) of random variable X are represented as $f_X(x)$ and $F_X(x)$, respectively. Expectation operator is denoted by $\mathbb{E}[\cdot]$. $Q(\cdot)$ represents the Gaussian-Q function.

[1]Note that due to imperfections in CSI and SIC, obtaining the closed-form expressions of sum ergodic rate is extremely difficult. We, therefore, use the Gaussian-Laguerre quadrature approximation. Next, we use logarithmic approximation to determine a simplified accurate closed-form expression of sum ergodic rate at each user in case of IBL transmission. In addition to this, under perfect CSI and SIC, the exact closed-form expression of ergodic sum rate is also presented.

[2]Due to involved mathematical complexity in deriving exact expressions for BLER and ergodic sum rate in FBL transmission, we first utilize Q-function approximation [37] and binomial approximation [38] and then derive highly accurate closed-form expressions for BLER and ergodic sum rate, respectively.

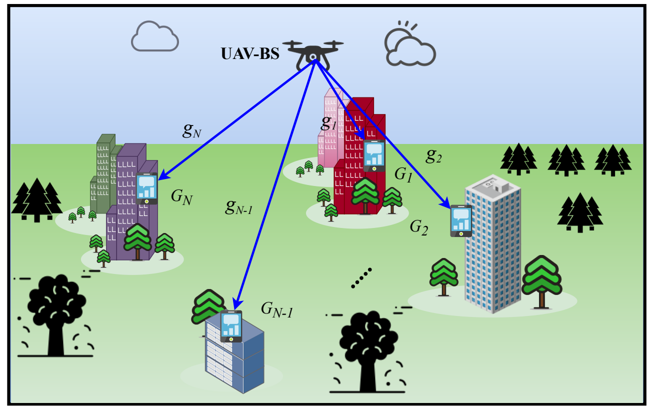

Fig. 1: UAV-BS aided Multiuser Communication using RSMA.

Organization of the Paper: The proposed system and RSMA-based signal are briefed in Section II. Section III presents the derived analytical expression of outage probability and ergodic sum rate considering IBL transmission. Section IV provides a detailed performance analysis of the FBL transmission regime. Section V discusses the sum rate maximization problem and its solution. Section VI discusses the simulation results, while Section VII concludes the paper.

II. System Model

We consider a downlink wireless network consisting of a UAV-BS that uses RSMA signaling to serve N GUs ($G_i \forall\, i\ \in \mathcal{N} \triangleq \{1,2,\cdots,N\}$), simultaneously. For analytical tractability, we have considered static UAV-BS that hovers at a fixed height and assume that all devices have a single antenna that operates in half-duplex mode. Further, we consider probabilistic LoS [39] fading model for the air to ground channel between UAV-BS and GUs. In particular, the channel gain between UAV-BS and G_i is given by

$$g_{_i} = \sqrt{\beta_{_i}} \times h_{_i}, \tag{1}$$

where $\beta_{_i} = \beta_o d_i^{-\mathfrak{a}_{_i}}$ represents the large scale pathloss with β_o denoting the reference channel gain per meter and d_i is the distance between UAV-BS and G_i, which is given by

$$d_i = \sqrt{(x_u - x_i)^2 + (y_u - y_i)^2 + (z_u - z_i)^2}, \tag{2}$$

where $\{x_u, y_u, z_u\}$ and $\{x_i, y_i, z_i\}$ are the 3-dimensional (3D) coordinates of UAV-BS and G_i, respectively. $\mathfrak{a}_{_i} = \left(\frac{\mathfrak{q}_1}{1+\mathfrak{q}_4 \exp(\mathfrak{q}_3(\psi_{_i} - \mathfrak{q}_4))} + \mathfrak{q}_2\right)$ denotes the pathloss exponent where $\mathfrak{q}_1$, $\mathfrak{q}_2$, $\mathfrak{q}_3$, and $\mathfrak{q}_4$ are constants defined in [39] and $\psi_{_i} = \arctan\left(\frac{z_{_i} - z_u}{\sqrt{(x_{_i} - x_u)^2 + (y_{_i} - y_u)^2}}\right)$. $h_{_i}$ is the small-scale fading and follows Nakagami-m distribution. The PDF and CDF of $|h_i|^2$ are given by

$$f_{|h_i|^2}(v) = \frac{m_i^{m_i} v^{m_i-1}}{\Omega_i^{m_i}\Gamma(m_i)}\exp\left(-\frac{m_i}{\Omega_i}v\right), \quad v \geq 0\,, \tag{3}$$

$$F_{|h_i|^2}(v) = \frac{1}{\Gamma(m_i)}\gamma\left(m_i, \frac{m_i v}{\Omega_i}\right), \quad v \geq 0\,, \tag{4}$$

where m_i and Ω_i are the shape and spread parameters. Unlike most of the work [26], [27], where channel estimation error is not considered, in this work, we consider the effect of channel estimation error h_{ei}. Thus, we can write $h_i = \hat{h}_i + h_{ei}$, where $\hat{h}_i$ is the estimated channel gain which also follows Nakagami-m distribution with parameters m_i and $\hat{\Omega}_i$. $h_{ei} \sim \mathcal{CN}(0, \sigma_{ei}^2)$ denotes the channel estimation error.

A. Downlink Transmission

As discussed earlier, UAV-BS uses RSMA signaling to transmit information to all users simultaneously. Thus, the message of each user has partitioned into two parts: the common message ($W_{i,c}$) and the private message ($W_{i,p}$). Next, the common messages of all the users are combined and encoded as a common message (s_c) using a common codebook available for each user. Further, the encoded private message of G_i is denoted by (s_i) [25]. s_c is the part of the message that is common between each user and will be decoded by each user. However, private message (s_i) is the message intended for G_i and will be decoded at G_i only. Therefore, transmit symbol at UAV-BS is given by

$$s = \sqrt{P_U}\left(\sqrt{a_c}s_c + \sum\nolimits_{i=1}^{N}\sqrt{a_i}s_i\right), \tag{5}$$

where P_U is the available transmit power at UAV-BS, a_c is the power coefficient allocated for transmitting common message, a_i is the power coefficient allocated to transmit private message to G_i and $a_c + \sum_{i=1}^{N} a_i = 1$. UAV-BS transmits the superimposed signal s to each user. The received signal at G_i is given by

$$\begin{aligned} y_i &= g_i s + n_i, \\ &= \underbrace{\sqrt{a_c\beta_i P_U}(\hat{h}_i + h_{ei})s_c}_{\text{desired common message}} + \underbrace{\sqrt{a_i\beta_i P_U}(\hat{h}_i + h_{ei})s_i}_{\text{desired private message}} \\ &\quad + \underbrace{\sum\nolimits_{j=1,i\neq j}^{N}\sqrt{a_j\beta_i P_U}(\hat{h}_i + h_{ei})s_j}_{\text{Interference}} + \underbrace{n_i}_{\text{AWGN}}, \end{aligned} \tag{6}$$

where $n_i \sim \mathcal{CN}(0, \sigma^2)$ represents the additive white Gaussian noise (AWGN) at G_i. Therefore, the signal to interference plus noise ratio (SINR) at G_i to decode s_c is given by

$$\gamma_{c,i} = \begin{cases} \dfrac{a_c\beta_i P_{\text{avg}}|\hat{h}_i|^2}{1+\beta_i P_{\text{avg}}|\hat{h}_i|^2(1-a_c)+\underbrace{\sigma_{ei}^2\beta_i P_{\text{avg}}}_{\text{imperfect-CSI}}}, & \text{imperfect CSI,} \\ \dfrac{a_c\beta_i P_{\text{avg}}|h_i|^2}{1+\beta_i P_{\text{avg}}|h_i|^2(1-a_c)}, & \text{perfect CSI,} \end{cases} \tag{7}$$

where $P_{\text{avg}} = P_U/\sigma^2$ is the transmit signal to noise ratio (SNR) at UAV-BS. After decoding s_c, each user applies SIC and to subtract s_c from the received message and thereafter decodes its private message. However, due to imperfection in CSI, there might be a slight misdetection of the common message. Unlike perfect SIC case, this misdetection might lead to error in SIC and there will be a interference which will be a function of a_c, β_i and P_{avg} [29], [30]. Thus, the corresponding SINR to decode the private message at G_i can be expressed as (8) given on top of next page.

Here, Ξ_i corresponds to the imperfections in SIC after decoding the common message with $\Xi_i = 0$ representing perfect SIC [29].

III. Infinite Blocklength Analysis

In this section, we consider the conventional IBL regime and analyze the performance of the considered system in terms of outage probability and achievable ergodic sum rate at each user.

A. Outage Probability

Since UAV-BS uses RSMA signaling to transmit the combination of common and private messages of all the users, each user first decodes the common message and then decodes its private message using SIC. Thus, the link between UAV-BS and each user can be considered to be in non-outage if and only if both messages are decoded successfully. In general, if either of the SINRs ($\gamma_{c,i}$ or $\gamma_{p,i}$) falls below the respective thresholds then outage will occur with a probability of

$$\mathrm{P_{Oi}} = 1 - \Pr\left(\gamma_{c,i} \geq \gamma_{th}^{c}, \gamma_{p,i} \geq \gamma_{th}^{p,i}\right), \tag{9}$$

where $\gamma_{th}^{c} = 2^{2R_c} - 1$ with a target rate of R_c and $\gamma_{th}^{p,i} = 2^{2R_{p,i}} - 1$ with a target rate of $R_{p,i}$.

Theorem 1. *The closed-form expression of outage probability* ($\mathrm{P_{Oi}}$) *at each user* ($\forall i \in \mathcal{N}$) *considering imperfect and perfect CSI and SIC is obtained as* (10) *on the top of next page.*

Proof. Substituting $\gamma_{c,i}$ and $\gamma_{p,i}$ for imperfect CSI and imperfect SIC from (7) and (8) into (9), we obtain Further applying algebraic simplifications in (11), we obtain

$$\begin{aligned} \mathrm{P_{Oi}} &= 1 - \Pr\left(|\hat{h}_i|^2 \geq \max\left(\hat{\gamma_{th}^{c}}, \hat{\gamma}_{th}^{p,i}\right)\right) \\ &= F_{|\hat{h}_i|^2}\left(\max\left(\hat{\gamma_{th}^{c}}, \hat{\gamma}_{th}^{p,i}\right)\right), \\ &= \int_0^{\max\left(\hat{\gamma_{th}^{c}}, \hat{\gamma}_{th}^{p,i}\right)} \frac{m_i^{m_i}v^{m_i-1}}{\Omega_i^{m_i}\Gamma(m_i)}\exp\left(-\frac{m_i}{\Omega_i}v\right)dv, \end{aligned} \tag{12}$$

where $\hat{\gamma_{th}^{c}} = \frac{(1+\sigma_{ei}^2\beta_i P_{\text{avg}})\gamma_{th}^{c}}{P_{\text{avg}}\beta_i\left(a_c-(1-a_c)\gamma_{th}^{c}\right)}$, $\hat{\gamma}_{th}^{p,i} = \frac{(1+\sigma_{ei}^2\beta_i P_{\text{avg}})\gamma_{th}^{p,i}}{P_{\text{avg}}\beta_i\left(a_i-(1+\Xi_i a_c-a_c-a_i)\gamma_{th}^{p,i}\right)}$. Solving (12) using [38, 3.381], we obtain (10). The outage probability at each user with perfect CSI and perfect SIC can be obtained by substituting $\sigma_{ei}^2 = 0$, $\Xi_i = 0$ and $\hat{\Omega}_i = \Omega_i$ in (12). ■

The sum throughput of the considered network can be evaluated as

$$\mathbb{T}_s = \sum\nolimits_{i=1}^{N}\mathbb{T}_i, \tag{13}$$

where $\mathbb{T}_i$ is the throughput at each user given by $\mathbb{T}_i = (1 - \mathrm{P_{Oi}})R_i/2$, where $R_i = R_{c,i} + R_{p,i}$. Now, we discuss the achievable ergodic sum rate at each user for both cases in the subsequent section.

$$\gamma_{p,i}=\begin{cases}\frac{a_i\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2}{1+\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2\sum\limits_{j=1,i\neq j}^{N}a_j+\underbrace{\sigma_{ei}^2\beta_i P_{\mathrm{avg}}}_{\text{imperfect-CSI}}+\underbrace{\Xi_i a_c\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2}_{\text{imperfect-SIC}}}, & \text{imperfect CSI and SIC},\\ \frac{a_i\beta_i P_{\mathrm{avg}}|h_i|^2}{1+\beta_i P_{\mathrm{avg}}|h_i|^2\sum\limits_{j=1,i\neq j}^{N}a_j}, & \text{perfect CSI and SIC}.\end{cases} \tag{8}$$

$$\mathrm{P_{Oi}}=\begin{cases}\text{imperfect}\begin{cases}\frac{1}{\Gamma(m_i)}\gamma\left(m_i,\frac{(1+\sigma_{ei}^2\beta_i P_{\mathrm{avg}})m_i\gamma_{th}^c}{P_{\mathrm{avg}}\beta_i\left(a_c-(1-a_c)\gamma_{th}^c\right)\hat{\Omega}_i}\right), & \text{if } \frac{a_c}{a_i}\leq\frac{\gamma_{th}^c(1+\gamma_{th}^{p,i})}{\gamma_{th}^{p,i}(1+\Xi_i\gamma_{th}^c)},\\ \frac{1}{\Gamma(m_i)}\gamma\left(m_i,\frac{(1+\sigma_{ei}^2\beta_i P_{\mathrm{avg}})m_i\gamma_{th}^{p,i}}{P_{\mathrm{avg}}\beta_i\left(a_i-(1+\Xi_i a_c-a_c-a_i)\gamma_{th}^{p,i}\right)\hat{\Omega}_i}\right), & \text{if } \frac{a_c}{a_i}>\frac{\gamma_{th}^c(1+\gamma_{th}^{p,i})}{\gamma_{th}^{p,i}(1+\Xi_i\gamma_{th}^c)},\end{cases}\\ \text{perfect}\begin{cases}\frac{1}{\Gamma(m_i)}\gamma\left(m_i,\frac{m_i\gamma_{th}^c}{P_{\mathrm{avg}}\beta_i\left(a_c-(1-a_c)\gamma_{th}^c\right)\Omega_i}\right), & \text{if } \frac{a_c}{a_i}\leq\frac{\gamma_{th}^c(1+\gamma_{th}^{p,i})}{\gamma_{th}^{p,i}},\\ \frac{1}{\Gamma(m_i)}\gamma\left(m_i,\frac{m_i\gamma_{th}^{p,i}}{P_{\mathrm{avg}}\beta_i\left(a_i-(1-a_c-a_i)\gamma_{th}^{p,i}\right)\Omega_i}\right), & \text{if } \frac{a_c}{a_i}>\frac{\gamma_{th}^c(1+\gamma_{th}^{p,i})}{\gamma_{th}^{p,i}}.\end{cases}\end{cases} \tag{10}$$

$$\mathrm{P_{Oi}}=1-\left(\frac{a_c\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2}{1+\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2(1-a_c)+\sigma_{ei}^2\beta_i P_{\mathrm{avg}}}\geq\gamma_{th}^c,\ \frac{a_i\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2}{1+\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2\sum\limits_{j=1,i\neq j}^{N}a_j+\sigma_{ei}^2\beta_i P_{\mathrm{avg}}+\Xi_i a_c\beta_i P_{\mathrm{avg}}|\hat{h}_i|^2}\geq\gamma_{th}^{p,i}\right). \tag{11}$$

B. Ergodic Sum Rate

The instantaneous rate at G_i for decoding a common message is given by

$$\hat{\zeta}_{c,i}=\log_2(1+\gamma_{c,i}), \tag{14}$$

where $\gamma_{c,i}$ is the SINR evaluated in (7). From (14), the achievable ergodic rate at G_i for decoding common message can be represented as

$$\zeta_{c,i}=\mathbb{E}\left[\hat{\zeta}_{c,i}\right]. \tag{15}$$

According to the principle of RSMA, the achievable ergodic rate at each user for decoding common message can not exceed ζ_c, where

$$\zeta_c=\min_{n\in\mathcal{N}}\zeta_{c,n}. \tag{16}$$

Further, ζ_c is distributed among all the users for their respective portion of the common message ($W_{i,c}$) satisfying the following condition

$$\sum\nolimits_{n\in\mathcal{N}}\tilde{\zeta}_{c,n}=\zeta_c=\min_{n\in\mathcal{N}}\zeta_{c,n}, \tag{17}$$

where $\tilde{\zeta}_{c,n}$ denotes the portion of ζ_c allocated to G_i. Further, the instantaneous rate at G_i for decoding private message is given by

$$\hat{\zeta}_{p,i}=\log_2(1+\gamma_{p,i}), \tag{18}$$

where $\gamma_{p,i}$ is the SINR evaluated in (8). From (18), the achievable ergodic rate at G_i for decoding private message can be represented as

$$\zeta_{p,i}=\mathbb{E}\left[\hat{\zeta}_{p,i}\right]. \tag{19}$$

The total rate at the G_i is given by

$$\zeta_i=(\tilde{\zeta}_{c,i}+\zeta_{p,i}). \tag{20}$$

The achievable sum rate of the considered network with the IBL transmission scheme is given by

$$\zeta_s=\sum\nolimits_{i\in\mathcal{N}}\zeta_i=\sum\nolimits_{i\in\mathcal{N}}(\tilde{\zeta}_{c,i}+\zeta_{p,i})=\zeta_c+\sum\nolimits_{i\in\mathcal{N}}\zeta_{p,i}. \tag{21}$$

Next, we present the analytical framework for solving (15) and (19) in order to obtain closed-form expression for ζ_s.

Theorem 2. *The achievable sum rate at each user after decoding common and private messages considering imperfect CSI and imperfect SIC is obtained as* (22) *on the top of next page.*

where p denotes the accuracy-complexity trade-off factor [40], t_k represents k^{th} root of the generalized Laguerre polynomial $L_p^{\varphi}(t)=\sum_{n=0}^{p}\binom{p+\varphi}{p-n}\frac{(-1)^n}{n!}t^n$ with $\varphi=m_i-1/2$. $w_k=\frac{t_k}{(p+1)^2[L_{p+1}^{\varphi}(t_k)]^2}$ represents the weight vector.

Proof. For the case of exposition we consider $X_i=|\hat{h}_i|^2$. The ergodic rate in (15) can be evaluated as

$$\zeta_{c,i}=\int_0^{\infty}\log_2(1+\gamma_{c,i})f_{X_i}(x)dx. \tag{23}$$

From (4) and (7), we can write

$$\begin{aligned}\zeta_{c,i}=\ &\frac{C_8^{m_i}}{\Gamma(m_i)}\int_0^{\infty}x^{m_i-1}\exp\left(-C_8x\right)\\ &\log_2\left(1+\frac{C_6x}{1+C_7x+C_{12}}\right)dx,\end{aligned} \tag{24}$$

where $C_6=a_c\beta_i P_{\mathrm{avg}}$, $C_7=(1-a_c)\beta_i P_{\mathrm{avg}}$, $C_8=m_i/\hat{\Omega}_i$, and $C_{12}=1+\sigma_{ei}^2\beta_i P_{\mathrm{avg}}$. Solving (24) is extremely difficult due to involved analytical complexity [3]. Therefore, we use

[3]Note that the presence of imperfect CSI and SIC errors makes it difficult to derive an exact closed-from expression of ergodic sum rate. However, with perfect CSI and SIC, an exact closed-form expression can be determined, which is given in Theorem 4.

$$\zeta_s=\sum_{k=0}^{p}w_k\left[\min_{n\in\{1,2,\cdots,N\}}\frac{m_n}{\hat{\Omega}_n\Gamma(m_n)}\log_2\left(1+\frac{a_c\hat{\Omega}_n\beta_nP_{\text{avg}}t_k}{m_n+\hat{\Omega}_n\beta_nP_{\text{avg}}k(1-a_c)+m_n\sigma_{en}^2\beta_nP_{\text{avg}}}\right)\right.$$
$$\left.+\sum_{i=1}^{N}\frac{m_i}{\hat{\Omega}_i\Gamma(m_i)}\log_2\left(1+\frac{a_i\beta_iP_{\text{avg}}t_k}{m_i+\hat{\Omega}_i\beta_iP_{\text{avg}}t_k\sum_{j=1,i\neq j}^{N}a_j+m_i\sigma_{ei}^2\beta_iP_{\text{avg}}+\hat{\Omega}_i\Xi_ia_c\beta_iP_{\text{avg}}t_k}\right)\right], \tag{22}$$

Gaussian-Laguerre quadrature [40] to obtain the solution of (24) as

$$\zeta_{c,i}=\frac{C_8}{\Gamma(m_i)}\sum_{k=0}^{p}w_k\log_2\left(1+\frac{(C_6/C_8)t_k}{1+(C_7/C_8)t_k+C_{12}}\right). \tag{25}$$

Similarly, we have

$$\zeta_{p,i}=\frac{C_8}{\Gamma(m_i)}\sum_{k=0}^{p}w_k\log_2\left(1+\frac{(C_9/C_8)t_k}{1+(C_{10}/C_8)t_k+C_{12}}\right). \tag{26}$$

where $C_9=a_i\beta_iP_{\text{avg}}$ and $C_{10}=(1+\Xi_ia_c-a_c-a_i)\beta_iP_{\text{avg}}$. Substituting (25) and (26) in (77), we obtain (22). ■

Although, (22) provides an exact expression of ζ_s considering imperfections in CSI and SIC estimates, the Gauss–Laguerre quadrature method involves high complexity due to which obtaining useful insights is not possible. We therefore use the approximation $\log(1+x)\simeq 2x/(2+x)$ [41] to obtain a much simpler yet accurate approximate solution to (24) in the following theorem.

Theorem 3. *An accurate closed-from expression of the ergodic sum rate at each user after decoding common and private messages considering imperfect CSI and SIC is given by* (27) *on the top of next page.*

Proof. Using the identity $\log(1+\gamma_{c,i})\simeq 2\gamma_{c,i}/(2+\gamma_{c,i})$ in (23), we have

$$\zeta_{c,i}\simeq\int_0^\infty\frac{2\gamma_{c,i}}{\ln(2)(2+\gamma_{c,i})}f_{X_i}(x)dx\,. \tag{28}$$

As shown in Fig. 2, there is a closer proximity between exact rate in (23) and approximate rate in (28). Substituting $\gamma_{c,i}$ from (7) into (28), $\zeta_{c,i}$ can be written as

$$\zeta_{c,i}\simeq\int_0^\infty\frac{a_c\beta_iP_{\text{avg}}x}{\begin{Bmatrix}\ln(2)(1+\beta_iP_{\text{avg}}x(1-a_c)\\+\sigma_{ei}^2\beta_iP_{\text{avg}}+a_c\beta_iP_{\text{avg}}x/2)\end{Bmatrix}}f_{X_i}(x)dx\,. \tag{29}$$

Substituting $f_{X_i}(x)$ from (3) and after performing some algebra, we obtain

$$\zeta_{c,i}\simeq\frac{C_6C_8^{m_i}}{\Gamma(m_i)\ln(2)(C_7+C_6/2)}\int_0^\infty\frac{x^{m_i}\exp(-C_8x)}{C_{12}(C_7+C_6/2)^{-1}+x}dx\,. \tag{30}$$

Solving for the above using [38, 3.387.10] and after some simplifications, we obtain

$$\zeta_{c,i}\simeq\frac{C_6m_i\exp\left(\frac{C_{12}C_8}{C_7+C_6/2}\right)\Gamma\left(-m_i,\frac{C_{12}C_8}{C_7+C_6/2}\right)}{(C_{12}C_8)^{-m_i}\ln(2)(C_7+C_6/2)^{m_i+1}}\,. \tag{31}$$

Further, the ergodic rate for decoding the private message in (19) can be evaluated as

$$\zeta_{p,i}=\int_0^\infty\log_2(1+\gamma_{p,i})f_{X_i}(x)dx. \tag{32}$$

Using the identity $\log(1+x)\simeq 2x/(2+x)$ [41], we obtain

$$\zeta_{p,i}\simeq\int_0^\infty\frac{2\gamma_{p,i}}{\ln(2)(2+\gamma_{p,i})}f_{X_i}(x)dx\,. \tag{33}$$

Substituting $\gamma_{p,i}$ from (8) into (33), $\zeta_{p,i}$ can be written as (34), given on next page.

From (3), (34), and after some algebra, we obtain

$$\zeta_{p,i}\simeq\frac{C_9C_8^{m_i}}{\Gamma(m_i)\ln(2)(C_{10}+C_9/2)}\int_0^\infty\frac{x^{m_i}\exp(-C_8x)}{C_{12}(C_{10}+C_9/2)^{-1}+x}dx\,, \tag{35}$$

Solving (35) using steps similar to (30), we have

$$\zeta_{p,i}\simeq\frac{C_9m_i\exp\left(\frac{C_{12}C_8}{C_{10}+C_9/2}\right)\Gamma\left(-m_i,\frac{C_{12}C_8}{C_{10}+C_9/2}\right)}{(C_{12}C_8)^{-m_i}\ln(2)(C_{10}+C_9/2)^{m_i+1}}. \tag{36}$$

Substituting (31) and (36) in (23), we obtain (27). ■

Evidently, (27) does not have any summation, thus its overall computational complexity is less compared to (22). Here we would like to mention that, the identity $\log(1+x)\simeq 2x/(2+x)$ is very accurate with error approaching zero at lower values of P_U. Although, for higher values of P_U, it provides a very close approximation with a maximum error of $\leq 5\%$. In other words, it provides a very close lower bound to the ζ_i, as shown in Fig 2. Note, that approximate analytical expression of the ergodic sum rate at each user after decoding common and private messages considering perfect CSI and perfect SIC can be obtained by substituting $\sigma_{ei}^2=0$, $\Xi_i=0$ and $\hat{\Omega}_i=\Omega_i$ in (27). In addition to this, we also provide an exact closed-form expression for the ergodic sum rate at each user in the following theorem.

Theorem 4. *The exact closed-form expression of the ergodic sum rate at each user after decoding common and private messages considering perfect CSI and perfect SIC is evaluated in* (37) *on the top of next page, where* $C_2=m_i/(\Omega_i\beta_iP_{\text{avg}})$, $C_3=m_i/(\Omega_iC_1)$, $C_1=(1-a_c)\beta_iP_{\text{avg}}$, $C_4=m_i/(\Omega_i(1-a_c)\beta_iP_{\text{avg}})$, $C_5=m_i/(\Omega_i(1-a_c-a_i)\beta_iP_{\text{avg}})$.

$$
\zeta_s \simeq \min_{n\in\{1,2,\cdots,N\}} \left[\frac{a_c m_n^{m_n+1}\exp\left(\frac{(1+\sigma_{en}^2\beta_n P_{\text{avg}})m_n}{\hat{\Omega}_n\beta_n P_{\text{avg}}(1-a_c/2)}\right)\Gamma\left(-m_n, \frac{(1+\sigma_{en}^2\beta_n P_{\text{avg}})m_n}{\hat{\Omega}_n\beta_n P_{\text{avg}}(1-a_c/2)}\right)}{(1+\sigma_{en}^2\beta_n P_{\text{avg}})^{-m_n}(\hat{\Omega}_n\beta_n P_{\text{avg}})^{m_n}\ln(2)(1-a_c/2)^{m_n+1}} \right]
$$
$$
+\sum_{i=1}^{N} \frac{a_i m_i^{m_i+1}\exp\left(\frac{(1+\sigma_{ei}^2\beta_i P_{\text{avg}})m_i}{\hat{\Omega}_i\beta_i P_{\text{avg}}(1+\Xi_i a_c-a_c-a_i/2)}\right)\Gamma\left(-m_i, \frac{(1+\sigma_{ei}^2\beta_i P_{\text{avg}})m_i}{\hat{\Omega}_i\beta_i P_{\text{avg}}(1+\Xi_i a_c-a_c-a_i/2)}\right)}{(1+\sigma_{ei}^2\beta_i P_{\text{avg}})^{-m_i}(\hat{\Omega}_i\beta_i P_{\text{avg}})^{m_i}\ln(2)(1+\Xi_i a_c-a_c-a_i/2)^{m_i+1}} \,. \quad (27)
$$

$$
\zeta_{p,i} = \int_0^\infty \frac{a_i\beta_i P_{\text{avg}}x}{\ln(2)(1+\beta_i P_{\text{avg}}x(1-a_c-a_i)+\sigma_{ei}^2\beta_i P_{\text{avg}}+\Xi_i a_c\beta_i P_{\text{avg}}x+a_i\beta_i P_{\text{avg}}x/2)} f_{X_i}(x)dx \,. \quad (34)
$$

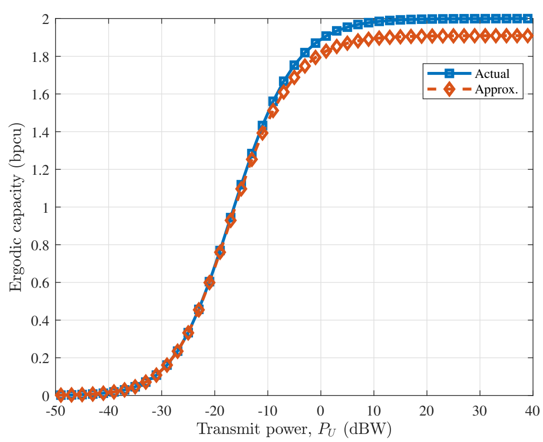

Fig. 2: Illustration of the accuracy of the logarithmic approximation of ergodic rate.

Proof. Substituting $\gamma_{c,i}$ from (7) for perfect CSI into (23) and after some mathematical simplifications, we have

$$
\zeta_{c,i} = \underbrace{\int_0^\infty \log_2(1+\beta_i P_{\text{avg}}x)f_{X_i}(x)dx}_{\mathrm{I}_1} - \underbrace{\int_0^\infty \log_2(1+C_1x)f_{X_i}(x)dx}_{\mathrm{I}_2} \,. \quad (38)
$$

Assuming $\beta_i P_{\text{avg}}x = y$, using f_{X_i} from (3) and solving for I_1, we obtain

$$
\mathrm{I}_1 = \frac{C_2^{m_i}}{\ln(2)\Gamma(m_i)}\int_0^\infty \ln(1+y)y^{m_i-1}\exp(-C_2y)dy. \quad (39)
$$

From [42], we have $\int_0^\infty \ln(1+t)t^{a-1}\exp(-bt)dt = \Gamma(a)\exp(b)\sum_{k=1}^{a}\frac{\Gamma(-a+k,b)}{b^k}$. Therefore, I_1 can be expressed as

$$
\mathrm{I}_1 = \frac{C_2^{m_i}\exp(C_2)}{\ln(2)}\sum_{k=1}^{m_i}\frac{\Gamma(-m_i+k,C_2)}{C_2^k}. \quad (40)
$$

Similarly, I_2 can be evaluated as

$$
\mathrm{I}_2 = \frac{C_3^{m_i}\exp(C_3)}{\ln(2)}\sum_{k=1}^{m_i}\frac{\Gamma(-m_i+k,C_3)}{C_3^k}. \quad (41)
$$

Using (40) and (41) into (38), we obtain

$$
\zeta_{c,i} = \frac{C_2^{m_i}}{\ln(2)}\sum_{k=1}^{m_i}\left(\frac{\exp(C_2)\Gamma(-m_i+k,C_2)}{C_2^k} - \frac{\exp(C_3)\Gamma(-m_i+k,C_3)}{(1-a_c)^{m_i}C_3^k}\right). \quad (42)
$$

We solve (32) similar to (38) and get

$$
\zeta_{p,i} = \frac{C_2^{m_i}}{\ln(2)}\sum_{k=1}^{m_i}\left(\frac{\exp(C_4)\Gamma(-m_i+k,C_4)}{(1-a_c)^{m_i}C_4^k} - \frac{\exp(C_5)\Gamma(-m_i+k,C_5)}{(1-a_c-a_i)^{m_i}C_5^k}\right). \quad (43)
$$

Substituting (42) and (43) on (77), we obtain (37). ∎

IV. Finite Blocklength Analysis

In this section, we analyze the reliability of the considered network in terms of BLER and achievable goodput at each user considering FBL transmission. UAV-BS transmits the short packet to each user formed by combining the common and private messages (see (5)).

A. Reliability Analysis

The instantaneous BLER at G_i for decoding a common message for sufficiently large blocklength (>100) can be approximated as

$$
\hat{\xi}_{c,i} \simeq Q\left(\frac{\hat{\zeta}_{c,i}-\rho_c}{\sqrt{\Lambda_{c,i}/r_c}}\right), \quad (44)
$$

where r_c is the blocklength of the common message, $\hat{\zeta}_{c,i}$ is the instantaneous rate at G_i given in (14), $\rho_c = \kappa_c/r_c$ with common message having κ_c number of information bits, and $\Lambda_{c,i}$ depicts the channel dispersion given by

$$
\Lambda_{c,i} = (\log_2 e)^2\left(1-\frac{1}{(1+\gamma_{c,i})^2}\right). \quad (45)
$$

From (44), the average BLER at G_i for decoding common message can be represented as

$$
\xi_{c,i} = \mathbb{E}\left[\hat{\xi}_{c,i}\right]. \quad (46)
$$

After decoding the common message, each user applies SIC and decodes its private message. Thus, the instantaneous

$$\zeta_s = \min_{i\in\mathcal{N}} \left[\frac{C_2^{m_i}}{\ln(2)}\sum_{k=1}^{m_i}\left(\frac{\Gamma(-m_i+k,C_2)}{\exp(-C_2)C_2^k} - \frac{\Gamma(-m_i+k,C_3)C_3^{-k}}{\exp(-C_3)(1-a_c)^{m_i}}\right)\right] + \sum_{i\in\mathcal{N}}\frac{C_2^{m_i}}{\ln(2)}\sum_{k=1}^{m_i}\left(\frac{\exp(C_4)\Gamma(-m_i+k,C_4)}{(1-a_c)^{m_i}C_4^k} - \frac{\exp(C_5)\Gamma(-m_i+k,C_5)}{(1-a_c-a_i)^{m_i}C_5^k}\right). \tag{37}$$

BLER at G_i while decoding a private message can be expressed as

$$\hat{\xi}_{p,i} \simeq Q\left(\frac{\hat{\zeta}_{p,i}-\rho_{p,i}}{\sqrt{\Lambda_{p,i}/r_{p,i}}}\right), \tag{47}$$

where $r_{p,i}$ is the blocklength of the private message of G_i, $\rho_{p,i} = \kappa_{p,i}/r_{p,i}$ with private message of G_i having $\kappa_{p,i}$ number of information bits, $\hat{\zeta}_{p,i}$ is the instantaneous rate for decoding private message at G_i given in (18), and $\Lambda_{p,i}$ depicts the channel dispersion given by

$$\Lambda_{p,i} = (\log_2 e)^2\left(1-\frac{1}{(1+\gamma_{p,i})^2}\right). \tag{48}$$

From (47), the average BLER at G_i for decoding private message can be represented as

$$\xi_{p,i} = \mathbb{E}\left[\hat{\xi}_{p,i}\right]. \tag{49}$$

The total average BLER at each user (ξ_i) will be the mean of individual BLERs of both common and private messages. Thus, from (46) and (49), we have

$$\xi_i = (\xi_{c,i}+\xi_{p,i})/2. \tag{50}$$

The closed-form expression of the average BLER at each user is derived in the following theorem.

Theorem 5. *The closed-form expression of the total average BLER at each user (ξ_i) with imperfect CSI and imperfect SIC is obtained as* (51)*, given on top next page.*

Proof. Substituting $\xi_{c,i}$ from (44) to (46), the average BLER at G_i for decoding common message can be expressed as

$$\xi_{c,i} \simeq \mathbb{E}\left[Q\left(\frac{\hat{\zeta}_{c,i}-\rho_c}{\sqrt{\Lambda_{c,i}/r_c}}\right)\right]. \tag{52}$$

Solving and finding a closed-form of (52) is analytically intractable. Thus, similar to [43], we can approximate $\xi_{c,i}$ as

$$\xi_{c,i} \simeq \int_0^\infty \Theta_{c,i}(x) f_{\gamma_{c,i}}(x)dx, \tag{53}$$

where $f_{\gamma_{c,i}}(\cdot)$ is the PDF of $\gamma_{c,i}$. $\Theta_{c,i}(\cdot)$ is an approximation used for $Q\left(\frac{\hat{\zeta}_{c,i}-\rho_c}{\sqrt{\Lambda_{c,i}/r_c}}\right)$ and can be expressed as

$$\Theta_{c,i}(\gamma_{c,i}) = \begin{cases} 1, & \gamma_{c,i}\le\vartheta_c,\\ 0.5-\lambda_c\sqrt{r_c}(\gamma_{c,i}-\theta_c), & \vartheta_c\le\gamma_{c,i}\le\nu_c,\\ 0, & \gamma_{c,i}\ge\nu_c,\end{cases} \tag{54}$$

where $\lambda_c = \frac{1}{2\pi\sqrt{2^{2\rho_c}-1}}$, $\theta_c = 2^{\rho_c}-1$, $\vartheta_c = \theta_c - \frac{1}{2\lambda_c\sqrt{r_c}}$, and $\nu_c = \theta_c + \frac{1}{2\lambda_c\sqrt{r_c}}$. The accuracy of linear approximation $\Theta_{c,i}$ with that of $Q\left(\frac{\hat{\zeta}_{c,i}-\rho_c}{\sqrt{\Lambda_{c,i}/r_c}}\right)$ is demonstrated in Fig. 3. It can be seen that the considered linear approximation of Q-function is very precise. Next, we use integration by parts and simplify (53) as

$$\xi_{c,i} \simeq \lambda_c\sqrt{r_c}\int_{\vartheta_c}^{\nu_c} F_{\gamma_{c,i}}(x)dx. \tag{55}$$

Next, similar to [43], the involved integration in the above equation can be solved using Riemann integral identity $\int_a^b g(y)dy = (b-a)g((a+b)/2)$. Thus, BLER in (55) can be expressed as

$$\xi_{c,i} \simeq \lambda_c\sqrt{r_c}(\nu_c-\vartheta_c)F_{\gamma_{c,i}}\left((\nu_c+\vartheta_c)/2\right). \tag{56}$$

To solve (56), we find the CDF $F_{\gamma_{c,i}}$ of the SINR $\gamma_{c,i}$ with imperfect CSI as

$$F_{\gamma_{c,i}}(x) = \Pr\left(\gamma_{c,i}<x\right). \tag{57}$$

From (7), (57) can be rewritten as

$$F_{\gamma_{c,i}}(x) = \Pr\left(\frac{a_c\beta_i P_{\text{avg}}|\hat{h}_i|^2}{\begin{Bmatrix}\beta_i P_{\text{avg}}|\hat{h}_i|^2(1-a_c)\\ +\sigma_{ei}^2\beta_i P_{\text{avg}}+1\end{Bmatrix}} < x\right). \tag{58}$$

After some mathematical manipulations, we have

$$\begin{aligned} F_{\gamma_{c,i}}(x) &= \Pr\left(|\hat{h}_i|^2 < \frac{C_{12}x}{\beta_i P_{\text{avg}}(a_c-(1-a_c)x)}\right)\\ &= F_{|\hat{h}_i|^2}\left(\frac{C_{12}x}{\beta_i P_{\text{avg}}(a_c-(1-a_c)x)}\right). \end{aligned} \tag{59}$$

Using (56) and (59), the BLER can be expressed as

$$\begin{aligned} \xi_{c,i} \simeq& \lambda_c\sqrt{r_c}(\nu_c-\vartheta_c)\\ &\times F_{|\hat{h}_i|^2}\left(\frac{C_{12}(\nu_c+\vartheta_c)}{\beta_i P_{\text{avg}}(2a_c-(1-a_c)(\nu_c+\vartheta_c))}\right). \end{aligned} \tag{60}$$

From (4) and (60), we obtain

$$\begin{aligned} \xi_{c,i} \simeq& (\lambda_c\sqrt{r_c}(\nu_c-\vartheta_c)/\Gamma(m_i))\\ &\times\gamma\left(m_i, \frac{C_{12}C_8(\nu_c+\vartheta_c)}{\beta_i P_{\text{avg}}(2a_c-(1-a_c)(\nu_c+\vartheta_c))}\right). \end{aligned} \tag{61}$$

Further, substituting $\xi_{p,i}$ from (47) to (49), the average BLER while decoding private message can be expressed as

$$\xi_{p,i} \simeq \mathbb{E}\left[Q\left(\frac{\hat{\zeta}_{p,i}-\rho_{p,i}}{\sqrt{\Lambda_{p,i}/r_{p,i}}}\right)\right]. \tag{62}$$

Further, similar to (53), we can write

$$\xi_{p,i} \simeq \int_0^\infty \Theta_{p,i}(x) f_{\gamma_{p,i}}(x)dx, \tag{63}$$

$$\xi_i \simeq \frac{1}{2\Gamma(m_i)}\left(\lambda_c\sqrt{r_c}(\nu_c-\vartheta_c)\gamma\left(m_i,\frac{(1+\sigma_{ei}^2\beta_iP_{\text{avg}})m_i(\nu_c+\vartheta_c)}{\hat{\Omega}_i\beta_iP_{\text{avg}}(2a_c-(1-a_c)(\nu_c+\vartheta_c))}\right)\right.$$
$$\left.+\lambda_{p,i}\sqrt{r_{p,i}}(\nu_{p,i}-\vartheta_{p,i})\gamma\left(m_i,\frac{(1+\sigma_{ei}^2\beta_iP_{\text{avg}})m_i(\nu_{p,i}+\vartheta_{p,i})}{\hat{\Omega}_i\beta_iP_{\text{avg}}(2a_i-(1+\Xi_ia_c-a_c-a_i)(\nu_{p,i}+\vartheta_{p,i}))}\right)\right). \quad (51)$$

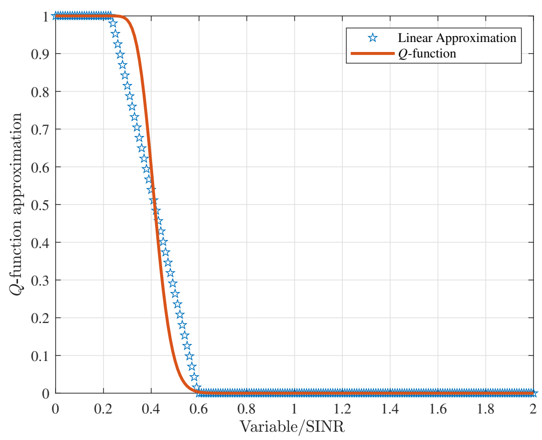

Fig. 3: Illustration of the accuracy of Q-function approximation.

where $f_{\gamma_{p,i}}(\cdot)$ is the PDF of $\gamma_{p,i}$. $\Theta_{p,i}(\cdot)$ is an approximation used for $Q\left(\frac{\hat{\zeta}_{p,i}-\rho_{p,i}}{\sqrt{\Lambda_{p,i}/r_{p,i}}}\right)$ and can be expressed as (64), given on the top of next page, where $\lambda_{p,i}=\frac{1}{2\pi\sqrt{2^{2\rho_{p,i}}-1}}$, $\theta_{p,i}=2^{\rho_{p,i}}-1$, $\vartheta_{p,i}=\theta_{p,i}-\frac{1}{2\lambda_{p,i}\sqrt{r_{p,i}}}$, and $\nu_{p,i}=\theta_{p,i}+\frac{1}{2\lambda_{p,i}\sqrt{r_{p,i}}}$. Similar to (56), BLER in (63) can be expressed as

$$\xi_{p,i}\simeq\lambda_{p,i}\sqrt{r_{p,i}}(\nu_{p,i}-\vartheta_{p,i})F_{\gamma_{p,i}}\left(\frac{\nu_{p,i}+\vartheta_{p,i}}{2}\right). \quad (65)$$

Now, the CDF $F_{\gamma_{p,i}}$ of the SINR $\gamma_{p,i}$ with imperfect CSI and imperfect SIC can be evaluated using

$$F_{\gamma_{p,i}}(x)=\Pr\left(\gamma_{p,i}<x\right). \quad (66)$$

From (7), (66) can be rewritten as (67) on the top of next page. After some mathematical manipulations, we have

$$F_{\gamma_{p,i}}(x)=\Pr\left(|\hat{h}_i|^2<\frac{(1+\sigma_{ei}^2\beta_iP_{\text{avg}})x}{\beta_iP_{\text{avg}}(a_i-(1+\Xi_ia_c-a_c-a_i)x)}\right),$$
$$=F_{|\hat{h}_i|^2}\left(\frac{(1+\sigma_{ei}^2\beta_iP_{\text{avg}})x}{\beta_iP_{\text{avg}}(a_i-(1+\Xi_ia_c-a_c-a_i)x)}\right). \quad (68)$$

From (3), (68) and (65), we obtain close-form of $\xi_{p,i}$ as

$$\xi_{p,i}\simeq(\lambda_{p,i}\sqrt{r_{p,i}}(\nu_{p,i}-\vartheta_{p,i})/\Gamma(m_i))$$
$$\gamma\left(m_i,\frac{C_8C_{12}(\nu_{p,i}+\vartheta_{p,i})}{\beta_iP_{\text{avg}}(2a_i-(1+\Xi_ia_c-a_c-a_i)(\nu_{p,i}+\vartheta_{p,i}))}\right). \quad (69)$$

Substituting (61) and (69) in (50), we obtain closed-form expression of average BLER at each user in (51). ■

Remark 1. *The average BLER at each user with perfect CSI and perfect SIC can be obtained by substituting $\sigma_{ei}^2=0$, $\Xi_i=0$ and $\hat{\Omega}_i=\Omega_i$ in (51).*

The overall BLER of the network is evaluated as the mean of all the average BLERs of all users and given by

$$\xi_s=\frac{1}{N}\sum_{i=1}^{N}\xi_i. \quad (70)$$

Further, the goodput at each user can be evaluated as

$$\mathbb{G}_i=\left(1-\frac{r_{lc,i}}{r_c}\right)\hat{R}_c(1-\tilde{\xi}_{c,i})+\left(1-\frac{r_{lp,i}}{r_{p,i}}\right)\hat{R}_{p,i}(1-\tilde{\xi}_{p,i}), \quad (71)$$

where $r_{lc,i}=r_c-\kappa_c$, $r_{lp,i}=r_{p,i}-\kappa_{p,i}$, $\hat{R}_c$ and $\hat{R}_{p,i}$ are target rate at each user. $\tilde{\xi}_{c,i}$ is obtained by substituting $\rho_c=e^{\kappa_c/r_c}-1$ in (61) and $\tilde{\xi}_{p,i}$ is obtained by substituting $\rho_{p,i}=e^{\kappa_{p,i}/r_{p,i}}-1$ in (69). The total achievable goodput of th network can be evaluated as

$$\mathbb{G}_s=\sum_{i=1}^{N}\mathbb{G}_i. \quad (72)$$

B. Achievable Rate

The instantaneous rate at G_i for decoding a common message considering FBL transmission is given by

$$\hat{\zeta}_{c,i}^{FB}=\hat{\zeta}_{c,i}-\sqrt{\Lambda_{c,i}/r_c}Q^{-1}(\tilde{\xi}_{c,i}), \quad (73)$$

where $\hat{\zeta}_{c,i}$ is given by (14) and $\Lambda_{c,i}$ and r_c are given in (44). $\tilde{\xi}_{c,i}$ is the constant maximum tolerable/desired BLER at G_i while decoding common message. The achievable ergodic rate at G_i for decoding common message can be evaluated by solving

$$\zeta_{c,i}^{FB}=\mathbb{E}\left[\hat{\zeta}_{c,i}^{FB}\right]. \quad (74)$$

Similarly, the instantaneous rate at G_i for decoding private message considering FBL transmission is given by

$$\hat{\zeta}_{p,i}^{FB}=\hat{\zeta}_{p,i}-\sqrt{\Lambda_{p,i}/r_{p,i}}Q^{-1}(\tilde{\xi}_{p,i}), \quad (75)$$

where $\hat{\zeta}_{p,i}$ is given by (18) and $\Lambda_{p,i}$ and $r_{p,i}$ are given in (47). $\tilde{\xi}_{p,i}$ is the constant tolerable BLER at G_i while decoding private message. The ergodic rate at G_i for decoding private message can be evaluated by solving

$$\zeta_{p,i}^{FB}=\mathbb{E}\left[\hat{\zeta}_{p,i}^{FB}\right]. \quad (76)$$

The achievable ergodic rate at G_i considering FBL regime can be expressed as

$$\zeta_i^{FB}=(\tilde{\zeta}_{c,i}^{FB}+\zeta_{p,i}^{FB}), \quad (77)$$

$$\Theta_{p,i}(\gamma_{p,i}) = \begin{cases} 1, & \gamma_{p,i} \leq \vartheta_{p,i}, \\ 0.5 - \lambda_{p,i}\sqrt{r_{p,i}}(\gamma_{p,i} - \theta_{p,i}), & \vartheta_{p,i} \leq \gamma_{p,i} \leq \nu_{p,i}, \\ 0, & \gamma_{p,i} \leq \nu_{p,i}. \end{cases} \tag{64}$$

$$F_{\gamma_{p,i}}(x) = \Pr\left(\frac{a_i\beta_i P_{\text{avg}}|\hat{h}_i|^2}{1 + \beta_i P_{\text{avg}}|\hat{h}_i|^2 \sum_{j=1,i\neq j}^{N} a_j + \sigma_{ei}^2\beta_i P_{\text{avg}} + \Xi_i a_c \beta_i P_{\text{avg}}|\hat{h}_i|^2} < x\right). \tag{67}$$

where $\tilde{\zeta}_{c,n}^{FB}$ denotes the portion of $\zeta_c^{FB} = \min_{n\in\mathcal{N}} \zeta_{c,n}^{FB}$ allocated to G_i. The achievable average sum rate of the considered network with FBL transmission scheme is given by

$$\begin{aligned} \zeta_s^{FB} &= \sum\nolimits_{i=1}^{N} \zeta_i^{FB} \\ &= \sum\nolimits_{i=1}^{N} (\tilde{\zeta}_{c,i}^{FB} + \zeta_{p,i}^{FB}) \\ &= \zeta_c^{FB} + \sum\nolimits_{i=1}^{N} \zeta_{p,i}^{FB}. \end{aligned} \tag{78}$$

Now, we solve (74) and (76) to obtain a closed-from expression of ζ_s^{FB} in the following theorem.

Theorem 6. *The achievable sum rate at each user after decoding common and private messages considering imperfect CSI and imperfect SIC is obtained as* (79) *on the top of next page.*

Proof. We derive (79) following steps similar to (22). ■

Further, Similar to (27), we derive a much simpler yet accurate approximate expression of ζ_s^{FB} in the following theorem.

Theorem 7. *The achievable ergodic sum rate at each user considering FBL transmission can be expressed as* (80)*, which given on next page.*

Proof. Substituting $\hat{\zeta}_{c,i}$ from (14), and $\Lambda_{c,i}$ and r_c from (44) into (73), we obtain

$$\begin{aligned} \zeta_{c,i}^{FB} = &\underbrace{\int_0^\infty \log_2(1+\gamma_{c,i}) f_{X_i}(x)dx}_{\mathrm{I}_3} \\ &\underbrace{- \int_0^\infty \frac{(\log_2 e)}{\sqrt{r_c}} \sqrt{(1-(1+\gamma_{c,i})^{-2})} Q^{-1}(\tilde{\xi}_{c,i}) f_{X_i}(x)dx}_{\mathrm{I}_4}. \end{aligned} \tag{81}$$

Note that integral I_3 is same as (23) which is solved in (27). Next, $\gamma_{c,i} > 0$ and $1/(1+\gamma_{c,i})^2 < 1$. Thus, we adopt binomial approximation $\sqrt{1-\frac{1}{(1+\gamma_{c,i})^2}} \simeq 1 - \frac{1}{2(1+\gamma_{c,i})^2}$ and obtain I_3 as

$$\mathrm{I}_4 \simeq \int_0^\infty C_{13}\left(1-(1/2)(1+\gamma_{c,i})^{-2}\right) f_{X_i}(x)dx, \tag{82}$$

where $C_{13} = \sqrt{(\log_2 e)^2/r_c} Q^{-1}(\tilde{\xi}_{c,i})$. Substituting $\gamma_{c,i}$ from (7) into (82) and after performing some algebraic manipulations, we obtain

$$\begin{aligned} \mathrm{I}_4 \simeq \int_0^\infty & C_{13}\left(1-\frac{1}{2}\left(1-\frac{C_6x}{(C_6+C_7)x+C_{12}}\right)^2\right) \\ & \times f_{X_i}(x)dx. \end{aligned} \tag{83}$$

In above $C_6x << (C_6+C_7)x + C_{12}$, thus using binomial approximation similar to (82), we obtain

$$\mathrm{I}_4 \simeq \int_0^\infty C_{13}\left(\frac{1}{2} + \frac{C_6x}{(C_6+C_7)x+C_{12}}\right) f_{X_i}(x)dx. \tag{84}$$

Using approach similar into (30), I_4 is obtained as

$$\mathrm{I}_4 \simeq \frac{C_{13}}{2} + \frac{C_{13}C_6 m_i \exp\left(\frac{C_8C_{12}}{C_6+C_7}\right)\Gamma\left(-m_i, \frac{C_8C_{12}}{C_6+C_7}\right)}{(C_8C_{12})^{-m_i}(C_6+C_7)^{m_i+1}}. \tag{85}$$

Substituting I_3 and I_4 in (81), $\zeta_{c,i}^{FB}$ is obtained as

$$\begin{aligned} \zeta_{c,i}^{FB} \quad &\simeq \zeta_{c,i} - \frac{C_{13}}{2} - \frac{C_{13}C_6 m_i (C_8C_{12})^{m_i}}{(C_6+C_7)^{m_i+1}} \\ &\times \exp\left(\frac{C_8C_{12}}{C_6+C_7}\right)\Gamma\left(-m_i, \frac{C_8C_{12}}{C_6+C_7}\right). \end{aligned} \tag{86}$$

Similarly, achievable ergodic rate at G_i after decoding private message considering FBL transmission can be evaluated as

$$\begin{aligned} \zeta_{p,i}^{FB} \quad &\simeq \zeta_{p,i} - \frac{C_{14}}{2} - \frac{C_{14}C_9 m_i (C_8C_{12})^{m_i}}{(C_9+C_{10})^{m_i+1}} \\ &\times \exp\left(\frac{C_8C_{12}}{C_9+C_{10}}\right)\Gamma\left(-m_i, \frac{C_8C_{12}}{C_9+C_{10}}\right), \end{aligned} \tag{87}$$

where $C_{14} = \sqrt{(\log_2 e)^2/r_{p,i}} Q^{-1}(\tilde{\xi}_{p,i})$ with $\zeta_{p,i}$ is given in (36). Substituting (86) and (87) into (78), we obtain (80). ■

Note that, due to involvement of RSMA in a FBL regime and to ensure the user fairness, the constraint of minimum required rate (common+private) at each user is levied. Additionally, limited flying zone of the UAV further imposes new challenges of resource management. Therefore, for efficient communication, a major problem is to provide optimum power allocation to each user at the UAV-BS and common rate distribution for each user, while guaranteeing the minimum QoS requirement at each user under the strict constraints of the limited UAV flying zone, the maximum tolerable BLER and achievable common rate.

Consequently, this imposes challenges to efficient resource management of the considered network. In particular, the major difficulty is to efficiently design the $\{a_c, a_i, \tilde{\zeta}_{c,i} \forall\ i\ \in \mathcal{N}\}$

$$\zeta_s^{FB} = \sum_{k=0}^{p} w_k \left[\min_{n\in\{1,2,\cdots,N\}} \frac{m_n}{\hat{\Omega}_n \Gamma(m_n)} \left\{ \log_2 \left(1 + \frac{a_c \hat{\Omega}_n \beta_n P_{\mathrm{avg}} t_k}{m_n + \hat{\Omega}_n \beta_n P_{\mathrm{avg}} k (1-a_c) + m_n \sigma_{en}^2 \beta_n P_{\mathrm{avg}}} \right) \right.\right.$$
$$\left. - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{c,i})}{\sqrt{r_c}} \sqrt{1 - \left(1 + \frac{a_c \hat{\Omega}_n \beta_n P_{\mathrm{avg}} t_k}{m_n + \hat{\Omega}_n \beta_n P_{\mathrm{avg}} k (1-a_c) + m_n \sigma_{en}^2 \beta_n P_{\mathrm{avg}}} \right)^{-2}} \right\} + \sum_{i=1}^{N} \frac{m_i}{\hat{\Omega}_i \Gamma(m_i)}$$
$$\times \left\{ \log_2 \left(1 + \frac{a_i \beta_i P_{\mathrm{avg}} t_k}{m_i + \hat{\Omega}_i \beta_i P_{\mathrm{avg}} t_k \sum_{j=1, i\neq j}^{N} a_j + m_i \sigma_{ei}^2 \beta_i P_{\mathrm{avg}} + \hat{\Omega}_i \Xi_i a_c \beta_i P_{\mathrm{avg}} t_k} \right) - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{c,i})}{\sqrt{r_c}} \right.$$
$$\left.\left. \times \sqrt{1 - \left(1 + \frac{a_i \beta_i P_{\mathrm{avg}} t_k}{m_i + \hat{\Omega}_i \beta_i P_{\mathrm{avg}} t_k \sum_{j=1, i\neq j}^{N} a_j + m_i \sigma_{ei}^2 \beta_i P_{\mathrm{avg}} + \hat{\Omega}_i \Xi_i a_c \beta_i P_{\mathrm{avg}} t_k} \right)^{-2}} \right\} \right]. \tag{79}$$

$$\zeta_s^{FB} \simeq \min_{n\in\mathcal{N}} \left[\frac{a_c m_n^{m_n+1} \exp\left(\frac{(1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}}) m_n}{\hat{\Omega}_n \beta_n P_{\mathrm{avg}} (1-a_c/2)} \right) \Gamma\left(-m_n, \frac{(1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}}) m_n}{\hat{\Omega}_n \beta_n P_{\mathrm{avg}} (1-a_c/2)} \right)}{(1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}})^{-m_n} (\hat{\Omega}_n \beta_n P_{\mathrm{avg}})^{m_n} \ln(2) (1-a_c/2)^{m_n+1}} \right.$$
$$\left. - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{c,n}) a_c m_n \exp\left(\frac{m_n (1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}})}{\hat{\Omega}_n \beta_n P_{\mathrm{avg}}} \right) \Gamma\left(-m_n, \frac{m_n (1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}})}{\hat{\Omega}_n \beta_n P_{\mathrm{avg}}} \right)}{2\sqrt{r_c} (1+\sigma_{en}^2 \beta_n P_{\mathrm{avg}})^{-m_n} (\hat{\Omega}_n \beta_n P_{\mathrm{avg}}/m_n)^{m_n}} - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{c,n})}{2\sqrt{r_c}} \right]$$
$$+ \sum_{i=1}^{N} \left[\frac{a_i m_i^{m_i+1} \exp\left(\frac{(1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}}) m_i}{\hat{\Omega}_i \beta_i P_{\mathrm{avg}} (1+\Xi_i a_c - a_c - a_i/2)} \right) \Gamma\left(-m_i, \frac{(1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}}) m_i}{\hat{\Omega}_i \beta_i P_{\mathrm{avg}} (1+\Xi_i a_c - a_c - a_i/2)} \right)}{(1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}})^{-m_i} (\hat{\Omega}_i \beta_i P_{\mathrm{avg}})^{m_i} \ln(2) (1+\Xi_i a_c - a_c - a_i/2)^{m_i+1}} - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{p,i})}{2\sqrt{r_{p,i}}} \right.$$
$$\left. - \frac{\log_2 e Q^{-1}(\tilde{\xi}_{p,i}) a_i m_i^{m_i+1} \exp\left(\frac{m_i (1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}})}{(1+\Xi_i a_c - a_c) \hat{\Omega}_i \beta_i P_{\mathrm{avg}}} \right) \Gamma\left(-m_i, \frac{m_i (1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}})}{(1+\Xi_i a_c - a_c) \hat{\Omega}_i \beta_i P_{\mathrm{avg}}} \right)}{2\sqrt{r_{p,i}} (1+\sigma_{ei}^2 \beta_i P_{\mathrm{avg}})^{-m_i} (1+\Xi_i a_c - a_c)^{m_i+1} (\hat{\Omega}_i \beta_i P_{\mathrm{avg}})^{m_i}} \right]. \tag{80}$$

at the UAV-BS and $\{\tilde{\xi}_{c,i}, \tilde{\xi}_{p,i}\}$ at each user while obtaining optimal $\{x_u, y_u, z_u\}$ to maximize the sum rate of the considered network subject to the above-discussed constraints. In what follows, with an aim to achieve these objectives, we formulate a sum rate maximization problem that jointly optimizes the aforementioned parameters in the following section.

V. Resource Allocation and UAV-BS Position Optimization

This section discusses the optimal resource allocation along with the position of UAV-BS to maximize the network's performance in terms of the sum rate with the FBL regime. The optimization problem to maximize the ergodic sum rate of the considered network can be formulated as

$$\text{(P0)} \quad \max_{\substack{\tilde{\xi}_{c,i}, \tilde{\xi}_{p,i}, x_u, y_u, z_u, \\ a_c, a_i, \tilde{\zeta}_{c,i} \forall\, i \in \mathcal{N}}} \left\{ \sum\nolimits_{i\in\mathcal{N}} \zeta_i^{FB} = \sum\nolimits_{i\in\mathcal{N}} (\tilde{\zeta}_{c,i}^{FB} + \zeta_{p,i}^{FB}) \right\}$$
$$\text{s.t.} \quad \tilde{\xi}_{c,i} \le \xi_{c,i,\max}, \tilde{\xi}_{p,i} \le \xi_{p,i,\max} \;\forall\, i \in \mathcal{N} \tag{88a}$$
$$a_c + \sum\nolimits_{i\in\mathcal{N}} a_i = 1, \tag{88b}$$
$$\sum\nolimits_{i\in\mathcal{N}} \tilde{\zeta}_{c,i}^{FB} \le \zeta_c^{FB} \tag{88c}$$
$$\tilde{\zeta}_{c,i}^{FB} + \zeta_{p,i}^{FB} \ge \zeta_{i,\min} \tag{88d}$$
$$x_{\min} \le x_u \le x_{\max}, y_{\min} \le y_u \le y_{\max},$$
$$z_{\min} \le z_u \le z_{\max}, \tag{88e}$$

where $(\cdot)_{\max}$ and $(\cdot)_{\min}$ denotes the maximum and minimum values of the respective parameters. (88a) corresponds to the maximum tolerable BLER constraint at each user. (88b) and (88c) are the allocated power coefficients and achievable common rate constraints discussed in earlier sections. The constraint (88d) ensures the minimum rate (common+private) requirement at each user. (88e) corresponds to the flying limitations of the UAV. From (80), it can be seen that due to coupling of variables and involvement of their ratio in the objective function (OF), problem (P0) is non-convex in nature. Therefore, it is extremely difficult to find a solution for problem (P0). Thus, considering the fact that $\tilde{\xi}_{c,i}$ and $\tilde{\xi}_{pi}$ are independent, we first obtain their optimal value in the following lemma:

Lemma 1. *The ζ_s^{FB} will be maximum w.r.t. $\tilde{\xi}_{c,i}$ and $\tilde{\xi}_{p,i}$ at the optimal value $\tilde{\xi}_{c,i}^o = \tilde{\xi}_{c,i,\max}$ and $\tilde{\xi}_{p,i}^o = \tilde{\xi}_{p,i,\max}$, respectively, for all $i \in \mathcal{N}$.*

Proof. To prove this lemma, we use contradiction method and assume that the optimal solutions $\tilde{\xi}_{c,i}^o$ and $\tilde{\xi}_{pi}^o \forall i \in \mathcal{N}$ satisfy strict inequality $\tilde{\xi}_{c,i} < \tilde{\xi}_{c,i,\max}$ and $\tilde{\xi}_{p,i} < \tilde{\xi}_{p,i,\max}$, respectively. With this assumption, in problem (P0), the values of $\tilde{\xi}_{c,i}^o$ and $\tilde{\xi}_{p,i}^o$ can be increased to achieve a higher sum rate while satisfying constraint (88a), hence, contradicting their optimality. ∎

It is note worthy that, similar to (88a), the objective function (OBF) of the problem (P0) will be maximum when $\sum_{i\in\mathcal{N}}\tilde{\zeta}_{c,i}^{FB} = \zeta_c^{FB}$ in constraint (88c). Therefore, the maximization of OBF in problem (P0) will be equivalent to the maximization of ζ_s^{FB} given in (80). Further, for simplicity, to make problem (P0) feasible, we assume that P_U is sufficiently high such that there exists at least a possible combination of the variables of the problem (P0) that satisfies rate constraints in (88d). Thus, the problem (P0) can be reformulated as

$$\text{(P1)} \quad \max_{x_u,y_u,z_u,a_c,a_i\ \forall\ i\in\mathcal{N}} \zeta_s^{FB} \quad \text{s.t.} \quad \text{(88b) and (88e) of (P0)}, \tag{89}$$

As evident from (80), due to involvement of channel dispersion and other constraints, problem (P1) is still non convex for obtained $\tilde{\xi}_{c,i}^{o}$ and $\tilde{\xi}_{p,i}^{o}\forall i\in\mathcal{N}$. Therefore, we adopt the alternating optimization (AO) approach and propose Algorithm 1 to solve each variable alternatively considering other variables constant. Specifically, we first fetch sub-optimal values of the power coefficients $\{a_c, a_1, a_2, \cdots, a_N\}$ for fixed $\{x_u, y_u, z_u\}$. These values are obtained from a sample space $\mathcal{P}_u$ which is a matrix of size $V\times(N+1)$ with V representing the total possible combination of values of $\{a_c, a_1, a_2, \cdots, a_N\}$ that satisfy (88b) i.e. summation of each row is equal to one. n^{th} column of $\mathcal{P}_u$ represents the values of a_{n-1} except the first column which represents the values of a_c. Afterwards, sub-optimal position of UAV is obtained by optimizing each coordinate of UAV iteratively, one by one over its minimum and maximum range, after fixing other two. Algorithm 1 repeats this process until the convergence is achieved i.e. the difference between two consecutive values of OBF of the problem (P1) falls below a threshold τ.

Finally, we determine the $\tilde{\zeta}_{c,i}^{FB}$ using a similar method that was used to obtain sub-optimal power coefficients. For the same, we simplify the constraint on the common rate distribution at each user as $\sum_{i\in\mathcal{N}}\alpha_i = 1$, where $\alpha_i = \tilde{\zeta}_{c,i}^{FB}/\zeta_c^{FB}\forall i\in\mathcal{N}$. Then we form a sample space $\mathcal{K}_c$ which is a matrix of size $D\times N$ with D representing the total possible combination of values of $\{\alpha_1, \alpha_2, \cdots, \alpha_N\}$ that satisfy (88d). Note that n^{th} column of $\mathcal{K}_c$ represents the values of α_n and sum of each row is equal to unity.

Complexity Analysis: Unlike brute force (BF)-method, Algorithm 1 uses sequential approach and find a sub-optimal value of each variable for a fixed value of other variables. Thus, it solves the problem (P1) with an approximate complexity of $L_{max}\mathcal{O}(V+C_x+C_y+C_z)$, where L_{max} is the maximum number of iteration required to solve the problem and C_k is the maximum of length of k with $k\in\{x,y,z\}$. Further, the computation complexity required to obtain $\tilde{\zeta}_{c,i}^{FB}\forall\ i\in\mathcal{N}$ is given by $\mathcal{O}(D)$. therefore, the problem (P0) is solved with a overall complexity of $L_{max}\mathcal{O}(V+C_x+C_y+C_z)+\mathcal{O}(D)$. Now, the complexity of BF-algorithm to solve the problem (P0) is approximately $\mathcal{O}(DVC_xC_yC_z)$ [19]. Therefore, considering $\mathcal{O}(DVC_xC_yC_z) >> L_{max}\mathcal{O}(V+C_x+C_y+C_z)+\mathcal{O}(D)$, the proposed method provides the solution similar to BF-method but with a much lesser complexity.

Remark 2. *The formulation and solution of the ergodic sum rate maximization problem for IBL will be similar to the FBL case, the details are omitted due to paucity of space.*

Algorithm 1 Alternating Optimization-based sub-optimal solution to Problem (P1)

Initialize $\tilde{\xi}_{p,i,\max}$, $\tilde{\xi}_{c,i,\max}$, P_U, $\mathcal{P}_u$, $l=0$, τ, $z_u=0$, $y_u=0$ and $x_u=0$
Initialize $x = x_{min}:1:x_{max}$, $z = z_{min}:1:z_{max}$, $y = y_{min}:1:y_{max}$,
Select $\{\alpha_1, \alpha_2, \cdots, \alpha_N\}$ from $\mathcal{K}_c$ randomly
repeat
  $l \to l+1$
  for $j = 1:V$ **do** $\{a_c, a_1, a_2, \cdots, a_N\} = \mathcal{P}_u(j,:)$
    Calculate $\text{temp}_0(j) = \zeta_s^{FB}$ using (80)
  end for
  Update $\{a_c, a_1, a_2, \cdots, a_N\} = \mathcal{P}_u\left(\arg\max_j(\text{temp}_0(j)),:\right)$
  for $p = 1:\text{length}(z)$ **do** $z_u = z(p)$
    Calculate $\text{temp}_1(p) = \zeta_s^{FB}$ using (80)
  end for
  Update $z_u = z\left(\arg\max_p(\text{temp}_1(p))\right)$
  for $q = 1:\text{length}(y)$ **do** $y_u = y(q)$
    Calculate $\text{temp}_2(q) = \zeta_s^{FB}$ using (80)
  end for
  Update $y_u = y\left(\arg\max_k(\text{temp}_2(q))\right)$
  for $r = 1:\text{length}(x)$ **do** $x_u = x(r)$
    Calculate $\text{temp}_3(r) = \zeta_s^{FB}$ using (80)
  end for
  Update $x_u = x\left(\arg\max_r(\text{temp}_3(r))\right)$
  Evaluate $\zeta_s^{FB}(l)$ using (80)
until $l>2$ and $\zeta_s^{FB}(l) - \zeta_s^{FB}(l-1) < \tau$

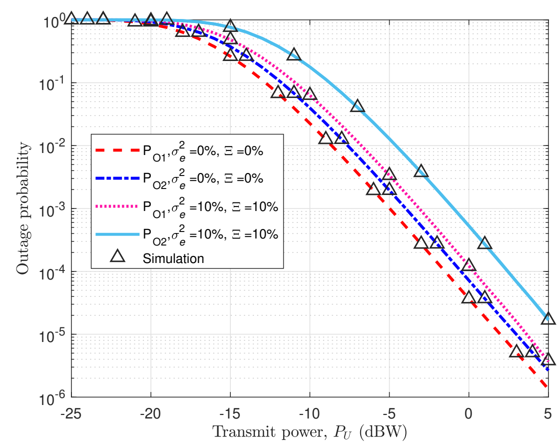

Fig. 4: Outage probability vs. transmit power

VI. Results and Discussions

This section validates the accuracy of the derived analytical expressions for IBL and FBL transmission schemes using extensive numerical simulations. Table II provides the considered simulation parameters. Without the loss of generality, we consider $N=2$, $\sigma_{e1}^2 = \sigma_{e2}^2 = \sigma_e^2$ and $\Xi_1 = \Xi_2 = \Xi$. However, all the analytical expressions and simulation results are applicable for any arbitrary N, Ξ_i and σ_{ei}^2 for all $i\in\mathcal{N}$. Moreover, assuming UAV-BS at the center, the coordinates of G_1 and G_2 are fixed at $(50,50,0)$m and $(10,50,0)$m, respectively [19].

Fig. 4 depicts the outage performance ($\mathrm{P_{O1}}$ and $\mathrm{P_{O2}}$) of

TABLE II: Simulation Parameters

Parameter	Value	Parameter	Value	Parameter	Value	Parameter	Value	Parameter	Value
σ^2	-110 dB	a_c	0.7	a_1	0.18	$\tilde{\xi}_{c,i}$	10^{-6}	$r_{p,i}\,\forall\, i$	250
a_2	0.12	$\Omega_i\,\forall\, i$	1	R_c	0.5 bpcu	$\tilde{\xi}_{p,i}$	10^{-6}	κ_c	75
$R_{p,i}\,\forall\, i$	0.15 bpcu	$m_i\,\forall\, i$	2	β_0	-30dB	$\kappa_{p,i}\,\forall\, i$	75	r_c	250

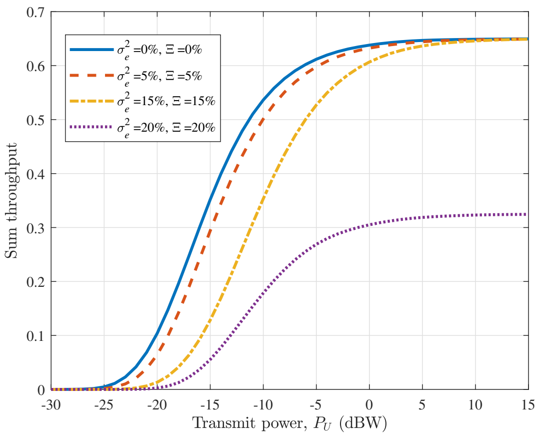

Fig. 5: Sum throughput vs. transmit power

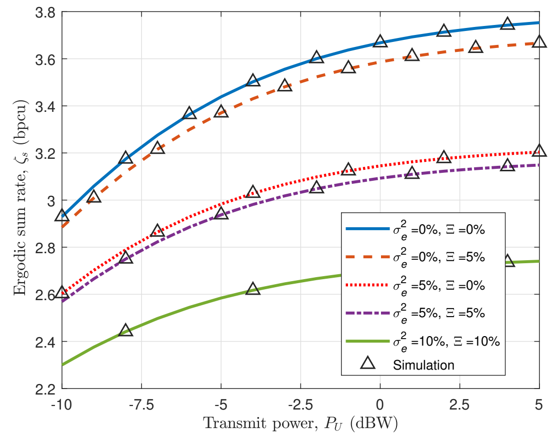

Fig. 6: Ergodic sum rate vs. transmit power

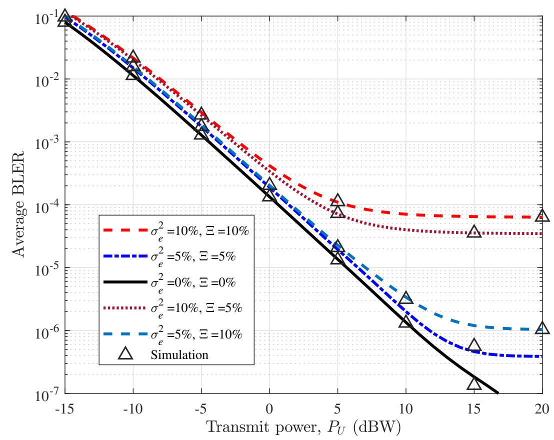

Fig. 7: Average BLER vs. transmit power at UAV-BS

each user with respect to (w.r.t.) transmit power used by UAV-BS. The close proximity between analytical and simulated curves validates the accuracy of the closed-from expression of outage probability obtained in (10). It can be observed that, due to an increase in SINR/SNR at each user, the outage probability decreases with increasing transmit power. Further, for the considered set of parameters, the outage performance of G_2 is better than that of G_1. In addition to this, we have also shown the impact of imperfect CSI and SIC. As expected, it can be seen that the estimation error in CSI and SIC leads to a significant degradation in the outage performance of each user.

In Fig. 5, we plot sum throughput ($\mathbb{T}_s$) versus transmitting power (P_U) for different settings of CSI and SIC errors. Clearly, the sum throughput is maximum when both CSI and SIC errors are not present. Even a low level of imperfection in CSI and SIC degrades the performance significantly. For example, when $\sigma_e^2 = 15\%$ and $\Xi = 15\%$, $\mathbb{T}_s \approx 0.52$ at $P_U = -5$dBW whereas it decreases to $\mathbb{T}_s \approx 0.28$ when both error are 20%. Therefore, with the increased SIC and CSI error, sum throughput degrades severely.

In Fig 6, the ergodic sum rate of the network is plotted w.r.t. transmit power for different percentages of estimation errors (σ_e^2, Ξ). The sum rate also improves with an increase in transmit power. Further, we also demonstrate the impact of estimation errors on the achievable ergodic sum rate. As obvious, the performance decreases with increase in percentage errors σ_e^2 and Ξ. However, the CSI error leads to significant degradation of the performance as compared to the SIC error. For example, there is an approximate decrease of 1 bits per channel use (bpcu) with $\sigma_e^2 = 5\%$ and $\Xi = 0\%$ whereas the decrease is less than 0.1 bpcu with $\sigma_e^2 = 0\%$ and $\Xi = 5\%$. Therefore, this highlights that the imperfection in CSI severely affects the ergodic rate than imperfection in SIC. The reason for this behavior is evident from the fact that imperfect SIC only affects the performance of each user while decoding the private message.

Fig 7 depicts the variation of achievable average BLER (ξ_s) w.r.t. transmit power used by UAV-BS for different percentage of estimation errors (σ_e^2, Ξ). One can observe the improvement in BLER with an increase in transmit power. The simulation result has proximity to the BLER obtained through derived closed-form expressions in (70). Additionally, the impact of estimation errors is also demonstrated in Fig. 7. It can be seen that the imperfection in CSI and SCI deteriorates the achievable BLER of the network. Also, with an increase in imperfection, the BLER achieves an early saturation. The reason for this behavior is that the increase in transmit power increases both the desired signal power as well as the

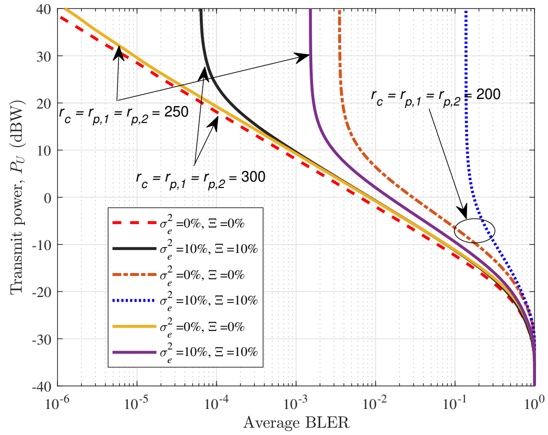

Fig. 8: Transmit power at UAV-BS vs. desired average BLER

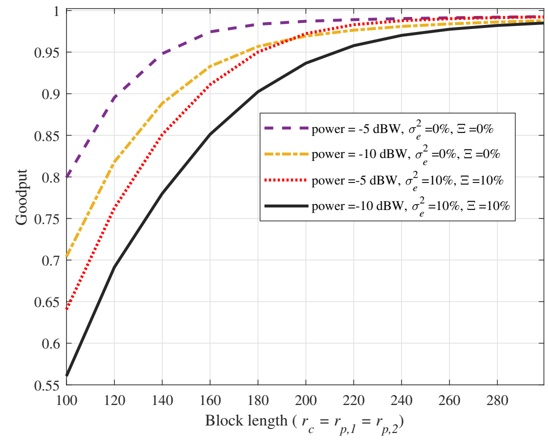

Fig. 10: Goodput vs blocklength

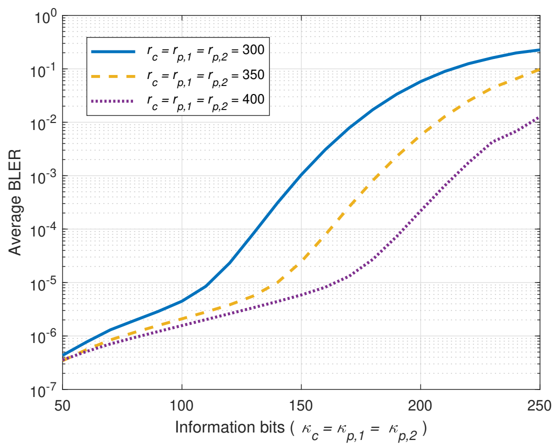

Fig. 9: Average BLER vs. information bits

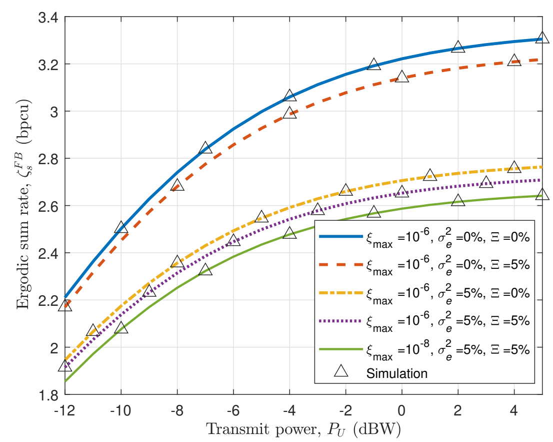

Fig. 11: Achievable ergodic sum rate vs. transmit power

interference power. Therefore, with the increase in CSI and SIC error, the interference further increases, causing an early saturation to the achievable BLER. For example, ξ_s saturates at 6×10^{-6} after $P_U > 15$dBW when $\sigma_e^2 = \Xi = 5\%$. However, for higher error $\sigma_e^2 = \Xi = 10\%$, this saturation is achieved at 2×10^{-4} after $P_U > 10$dBW.

In Fig. 8, we discuss the required transmit power (P_U) for obtaining a desired average BLER (ξ_s) at the network. It is obtained by plotting P_U w.r.t. (ξ_s) for different estimation errors and blocklength (r_c, $r_{p,1}$ and $r_{p,2}$). With increase in (ξ_s), the required transmit power P_U decreases. It can also be seen that an increase in blocklength leads to a reduction in net transmit power required to achieve a fixed BLER. Further, performance degrades with an increase in the imperfection of CSI and SIC at each user.

Next, we analyze the impact of number of information bits (κ_c, $\kappa_{p,1}$, and $\kappa_{p,2}$) on the achievable BLER (ξ_s) of the network in Fig 9. It is obtained by plotting ξ_s w.r.t. $\kappa_c = \kappa_{p,1} = \kappa_{p,2}$ with $P_U = 10$dBW for three different values of blocklength. It can be observed that for larger information bits, the BLER is also high. This observation can be verified by the fact that an increase in the number of bits also increases error probability. Thus, there exists a trade-off between total information bits and achievable BLER of the system. In addition to this, the impact of blocklength on ξ_s is also shown in Fig 9. It can be observed there is a significant improvement in the performance with an increase in blocklength. Further, we show achieve able goodput ($\mathbb{G}_s$) of the network in Fig. 10. It is obtained by plotting $\mathbb{G}_s$ w.r.t. total blocklength for different transmit power. It can be seen that with an increase in blocklength, the goodput also increases. In addition to this, as obvious, the decrease in transmit power leads to significant performance degradation. Further, goodput decreases with an increase in the imperfection of CSI and SIC at each user.

Fig. 11 depicts the plot of achievable ergodic sum rate (ζ_s^{FB}) of the network versus transmit power (P_U) and demonstrates the impact of maximum tolerable BLER $\hat{\xi}_{c,i} = \hat{\xi}_{p,i} = \xi_{\max}$. Simulation results clearly validates the accuracy of the analytical expression obtained in (78). It can observed that with decrease in $\xi_{\max}$, ζ_s^{FB} also decreases. Further, ζ_s^{FB} also degrades with increase in percentage error of CSI and SIC. However, the channel estimation error leads to a significant degradation of the performance as compared to the SIC error. For example, there is an approximate decrease of 0.8 bpcu with $\sigma_e^2 = 5\%$ and $\Xi = 0\%$ where as the decrease is less

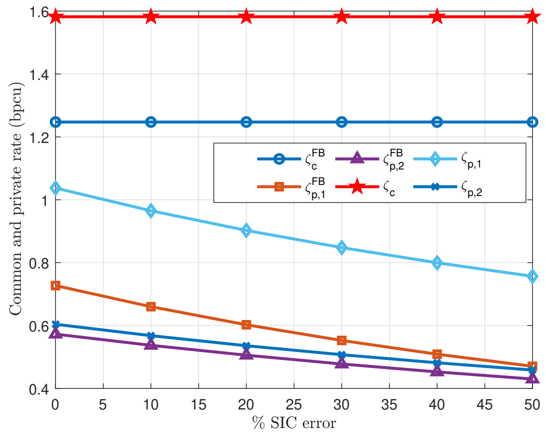

Fig. 12: Impact of Imperfect SIC

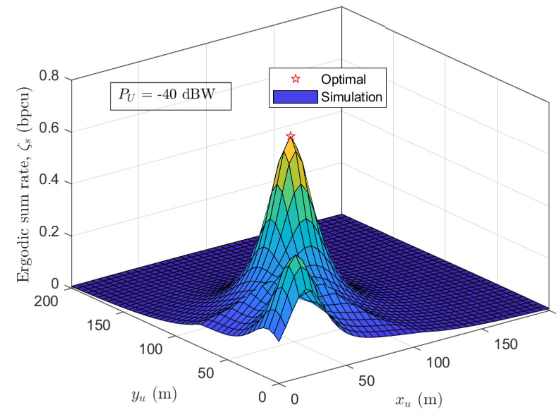

Fig. 14: Optimal UAV-BS position for IBL at $P_U = -40$ dBW

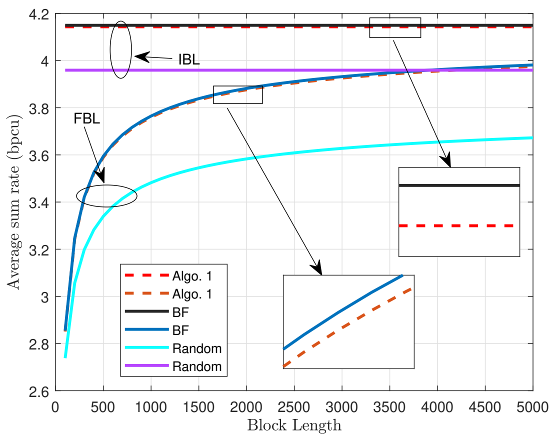

Fig. 13: Infinite and finite blocklength transmisison

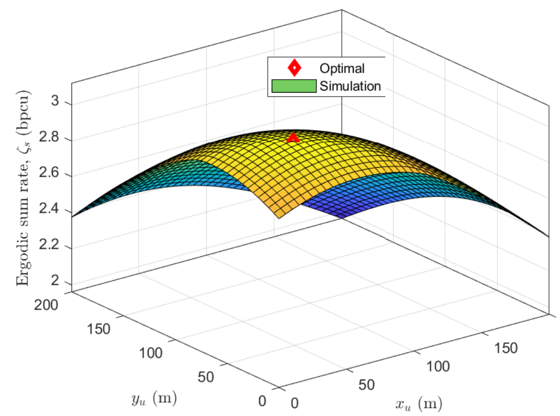

Fig. 15: Optimal UAV-BS position for IBL at $P_U = -5$ dBW

than 0.1 bpcu with $\sigma_e^2 = 0\%$ and $\Xi = 5\%$. Therefore, this highlights that the imperfection in CSI severely effects the achievable ergodic rate than imperfection in SIC. The reason for this behavior is evident from the fact that imperfect SIC only effects the performance of each user while decoding the private message.

In Fig. 12, we demonstrates the impact of percentage SIC error on the performance of each user for IBL and FBL transmission schemes. It depicts the plot of ergodic rates ζ_c, ζ_c^{FB}, $\zeta_{p,1}$, $\zeta_{p,1}^{FB}$, $\zeta_{p,2}$, and $\zeta_{p,2}^{FB}$ versus percentage SIC error at $P_U = 10$ dBW and $\sigma_e^2 = 10\%$. It can be observed that SIC error has no effect on the achievable ergodic rate after decoding common message at each user for each transmission regime. However, SIC error only degrades the performance corresponding to the decoding of private message at each user, as SIC is performed after decoding common message.

Fig 13 compares the performance of the network considering IBL and FBL regimes for three different cases: 1) BF, 2) Proposed, and 3) Random. It is obtained by plotting ζ_s and ζ_s^{FB} versus blocklength $r_c = r_{p,1} = r_{p,2}$ with $\Xi = \sigma_e^2 = 5\%$, $P_U = -5$ dBW, $\xi_{\max} = 10^{-8}$, $\{x_1, y_1, z_1\} = \{80, 70, 0\}$ m, and $\{x_2, y_2, z_2\} = \{10, 20, 0\}$ m. As expected the performance in the case of IBL regime is better than that of FBL

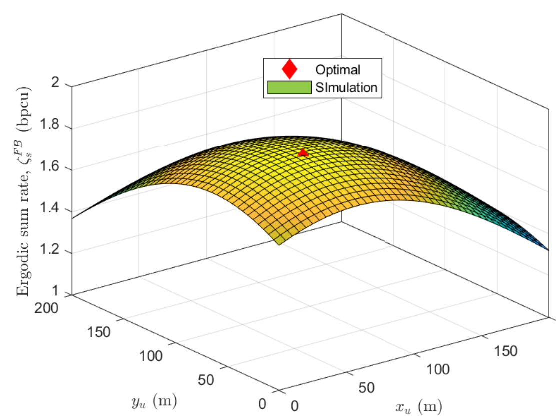

Fig. 16: Optimal UAV-BS position for FBL at $P_U = -5$ dBW

regime for all three cases. However, for the considered set of parameters, the proposed method provides approximately $1\% - 8\%$ better performance compared to the random case for both the regimes. Also, the performance of proposed method is at par with BF case with a difference of less than 1% throughout the range of blocklength, hence confirming the

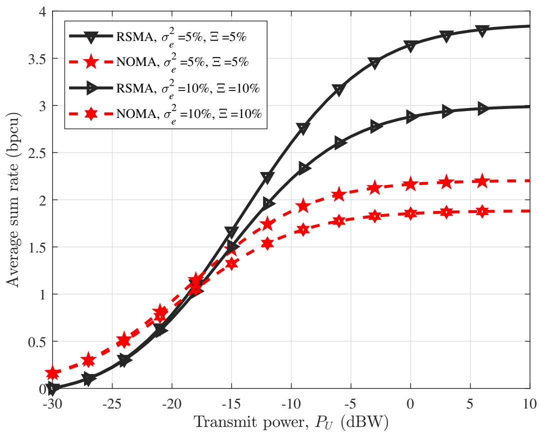

Fig. 17: RSMA vs. NOMA

efficacy of the proposed method. Additionally, one can also observe that the ζ_s^{FB} approaches ζ_s as blocklength increases. Thus, we can conclude that $\zeta_s^{FB} \approx \zeta_s$ when blocklength is very large (or approaches infinity) for all three cases. Next, we demonstrate the accuracy of the Algorithm 1 for obtaining the optimal position of the UAV-BS. For the same, after fixing the height of the UAV-BS at 10m, we plot ζ_s versus in x and y coordinates of UAV-BS in Fig. 14 and Fig. 15 for $P_U = -40$ dBW and $P_U = -5$ dBW, respectively, and ζ_s^{FB} versus in x and y coordinates of UAV-BS in Fig. 16 for $P_U = -5$ dBW. It can be seen that there is a close proximity between the optimal coordinates obtained using Algorithm 1 and simulations.

In Fig 17, we plot and compare the achievable sum rate of the proposed network considering RSMA signaling with that of NOMA signaling. For NOMA, we substitute a_c=0 and s_c=0 in (5), the transmit symbol from UAV-BS with NOMA is given by $s_{NOMA} = \sum_{i=0}^{N} \sqrt{(a_i P_U)} s_i$. The rate for NOMA is obtained by substituting s_{NOMA} instead s in (5) and performing multi-step SIC. Also, optimal power allocated to each user in NOMA is obtained by replacing ζ_s^{FB} by sum rate of NOMA signalling ($\sum_{i \in \mathcal{N}} \zeta_i^{NOMA}$) in Problem (P1). It can be observed that performance with both the schemes increases with an increase in power. However, the performance with NOMA is slightly better at the low SNRs, while RSMA signaling dominates at high SNRs. As in the case of NOMA, an increase in transmit power results in an increase in signal power as well as the interference hence resulting in a limited improvement to the performance. Further, the impact of imperfection in CSI and SIC is also discussed in Fig 17. Similar to RSMA, the degradation of the performance with NOMA signaling can be clearly seen. However, the noteworthy point is that even after degradation in performance, RSMA dominates NOMA with a significant difference in ergodic sum rate at higher power. This highlights the robustness of RSMA in the presence of imperfect CSI and SIC as compared to NOMA. Thus, the use of RSMA is justified instead of NOMA in such scenarios. Besides, NOMA is very complex since the strong user has to decode the message of all other users and therefore requires multiple layers of SIC. In contrast, RSMA can get better performance than NOMA but with a single-stage SIC, therefore significantly reducing the receiver complexity.

VII. Conclusion

In this work, the performance of a downlink wireless network consisting of a UAV-BS that used RSMA signaling to serve multiple GUs simultaneously was investigated under IBL and FBL transmission regimes. Further, considering the near to practical scenario, the impact of imperfect CSI and SIC on the network's performance was investigated. Moreover, the closed-form expressions of outage probability and achievable ergodic rate for IBL transmission, and BLER and achievable ergodic rate for FBL transmission were derived. We jointly optimized the 3D-position of the UAV, power allocated to each user and common rate distribution at each user using an AO-based iterative algorithm subject to the practical constraints such as maximum tolerable BLER, minimum private and total rate at each user under the imperfect SIC and CSI estimates. Numerical results demonstrated the impact of these imperfections on the performance metrics for both the transmission regimes. It is concluded that the imperfect CSI provides more severe degradation on the performance than imperfect SIC. Further, the trade-off between transmit power and achievable BLER was also highlighted. Finally, the effectiveness of the use of RSMA signaling as compared to NOMA in such scenarios was demonstrated and we found that performance with RSMA and NOMA was comparable at the lower range of SNR, while achievable ergodic rate using RSMA signaling dominated with that of NOMA signaling throughout the range of SNR.

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